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Review

Exploring Crossmodal Associations Between Sound and the Chemical Senses: A Systematic Review Including Interactive Visualizations

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Abstract

This is the first systematic review that focuses on the influence of product-intrinsic and extrinsic sounds on the chemical senses involving both food and aroma stimuli. This review has a particular focus on all methodological details (stimuli, experimental design, dependent variables, and data analysis techniques) of 95 experiments, published in 83 publications from 2012 to 2023. 329 distinct crossmodal auditory–chemosensory associations were uncovered across this analysis. What is more, instead of relying solely on static figures and tables, we created a first-of-its-kind comprehensive Power BI dashboard (interactive data visualization tool by Microsoft) on methodologies and significant findings, incorporating various filters and visualizations allowing readers to explore statistics for specific subsets of experiments. We believe that this review can be helpful for researchers and practitioners working in the food and beverage industry and beyond these scopes (e.g., cosmetics). Theoretical and practical implications discussed in this article point to computational approaches that facilitate decision-making regarding multisensory experimental methodology design.

Keywords

auditory, crossmodal, experience, flavor, food, music, aroma, visualization

1. Introduction

Numerous studies have demonstrated that auditory input significantly impacts gustatory and olfactory perception (i.e., chemosensory experience), regardless of whether individuals are conscious of this connection (Spence, 2014; Spence *et al.*, 2019). This phenomenon is commonly approached *via* the study of crossmodal associations, which are the connections that many individuals tend to experience between features, attributes, or dimensions of the different sensory inputs, e.g., between audition and the chemical senses (Spence, 2011; Parise and Spence, 2013). In fact, the role of what we hear in the experience of the chemical senses is being further considered in multisensory-oriented research (Spence, 2019a).

Since the food, beverage, and to a lesser extent, fragrance industries are becoming increasingly aware of these associations, the auditory experiences surrounding such products (e.g., packaging and advertising) are being customized to modulate sensory perceptions toward enhancing olfactory and gustatory experiences. For instance, when a customer is selecting a product, sounds can modulate sensory expectations regarding that product (Velasco *et al.*, 2014a; Spence and Wang, 2015; Wang and Spence, 2018a; Hwang *et al.*, 2020). In the context of product usage, sounds can also influence the actual product experience and evaluation. These sounds capable of influencing such sensory experiences are typically categorized as either intrinsic or extrinsic to the product.

For the present work, we followed that categorization by framing sounds from those two different perspectives (product-intrinsic *vs* extrinsic sounds). Product-intrinsic sounds were defined as the sounds directly associated with a product or its packaging (e.g., the spray sound of a perfume, the crunchiness of food when being chewed, or the sound of the sizzle of steak on a hotplate). Conversely, product-extrinsic sounds were defined as those not inherently associated with the product (e.g., background noise or music; see Spence *et al.*, 2019 for further support on such definitions).

Crossmodal associations between audition and the chemical senses appear to have been studied primarily in the food and beverage sector, addressing the fundamental question of the influence of sound on flavor expectation, perception, and evaluation. A few narrative reviews have assessed some of the empirical effects of sounds on flavor perception, feeding behavior, and olfaction, covering background noise/music, product-intrinsic sounds, packaging sounds, and offline/online sonic experiences (Knöferle and Spence, 2012; Spence, 2012, 2021; Wang and Spence, 2018a; Spence *et al.*, 2019, 2021). However, the literature still lacks a structured characterization of methodological aspects, including stimuli, experimental design, dependent variables, and

data analysis techniques of the most recent studies in this field (i.e., systematic review; Poklepović Peričić and Tanveer, 2019).

A recent systematic review scrutinized the body of work on the cross-modal associations between audition and taste, while characterizing the sample, setting, auditory stimuli, taste stimuli, measures, and experimental design (Guedes *et al.*, 2023b). However, these findings were confined to the five basic tastes (sweet, bitter, salty, sour, and umami), which are specifically detected by taste receptors. Therefore, there is an opportunity for an updated systematization in the field that includes flavors (i.e., perceptions formed by the interaction of all senses when tasting food/beverage; Auvray and Spence, 2008) and potentially goes beyond food and beverage stimuli.

Given the preceding considerations, this report proposes the first systematic review that focuses on the influence of product-intrinsic and extrinsic sounds on the chemical senses involving both food and aroma stimuli. Table 1 provides a comparative analysis of the characteristics of the previous reviews and the present work, underscoring the appropriateness of the current study. This systematic review aims to address such a gap in the literature by rendering a more comprehensive and up-to-date overview of studies.

This review seeks to characterize the methodological attributes (auditory stimuli, chemosensory stimuli, participants, type of setting, experimental condition design, measures, and data analysis) of these experimental studies, disclosing and quantifying (in percentage) the prevalent types within each attribute. This comparative analysis will assist researchers in visualizing the most used types for each attribute, foreseeing new discoveries in experimental methodology design and computational applications. Moreover, the present

Table 1.

Comparison between the characteristics of the previous reviews and the present

Reviews	Systematic methodo- logy	Flavors beyond tastes	Product- intrinsic sounds	Product- extrinsic sounds	Aromas	Methods characteri- zation	Interactive visuali- zation
Spence (2012)		✓	✓	✓	✓		
Knöferle and Spence (2012)		✓		✓			
Wang and Spence (2018a)		✓	✓				
Spence <i>et al.</i> (2019)		✓		✓	✓	✓	
Spence <i>et al.</i> (2021)		✓		✓	✓	✓	
Spence (2021)		✓		✓	✓	✓	
Guedes <i>et al.</i> (2023b)	✓			✓		✓	
The present review	✓	✓	✓	✓	✓	✓	✓

study addresses not only actual tasting/smelling but also expectations preceding the actual experience, in which sounds play a crucial role. On top of that, it transcends the conventional approach of systematic reviews, which rely solely on static figures and tables. Instead, this study introduces an innovative approach by offering a comprehensive interactive Power BI dashboard (an interactive data visualization tool by Microsoft) on methodologies and findings, incorporating various filters and visualizations that allow readers to explore the statistical data for specific subsets of experiments.

In sum, as particular innovative contributions compared to previous similar studies, auditory and chemosensory stimuli are discussed separately from the other methodological attributes to categorize and compare these stimuli in more detail. Second, associations between sound parameters and chemosensory experiences in the selected studies are compiled to investigate potential patterns and trends in these associations. These parameters and experiences are further disaggregated into comparable low-level parameters, with modulations on perceptions and personal preferences analyzed separately, enabling quantification of each association. Third, the obtained findings being placed in interactive visualizations facilitate the exploration of prospective opportunities for innovating the currently used methodology by researchers and practitioners in this area. Finally, this work focuses on chemosensory experiences measured *via* both food stimuli (i.e., foods and beverages) and aromas.

2. Systematic Methodology

The present systematic review adhered to the PRISMA guidelines proposed by Moher *et al.* (2009) while utilizing the Parsifal (see Note 1) web-based application. Moreover, Power BI software was used to create a complete interactive dashboard to facilitate readers to explore the results in multiple ways (for interactive engagement, follow the link <https://bit.ly/visaudchexps>).

2.1. Research Questions

In line with our primary objective, the following research questions were formulated, aiming at elucidating the current literature on the auditory–chemosensory relationship:

- RQ1: Which methods and materials are currently used to evaluate the influence of sounds on the experience of the chemical senses?
- RQ2: Which auditory parameters have been found to influence chemosensory experiences?

2.2. Information Sources and Search Strategy

To answer these research questions, an exhaustive bibliographic search of peer-reviewed articles containing experimental studies indexed in Scopus

(Note 2), Web of Science (Note 3), SAGEPub (Note 4), Emerald (Note 5, and MDPI (Note 6) was performed. These databases were chosen because they offer a wide variety of articles and include relevant and prestigious journals in the field of study. Adapted to each database, the following search string was used to conduct the review:

Title/Keywords/Abstract contains [(flavor OR taste OR tasting OR olfactory OR odor OR aroma OR smell) AND (sound OR auditory OR sonic OR music OR musical OR soundscape) AND (perception OR experience OR consumer OR marketing OR multisensory OR crossmodal)] AND All fields contain (consumer OR product)

The bibliographic search was conducted in May 2022.

2.3. Eligibility Criteria and Selection Process

Eligible studies evaluated at least one auditory stimulus intending to correspond to chemosensory experiences. In particular, the primary inclusion criteria were as follows:

- The paper was published after 2012.
- The paper was written in English.
- The paper constituted a scientific journal article, book chapter, or conference proceeding.

Note that the minimum year of publication was set to 2012 because the last comprehensive review that included flavors, aromas, product-intrinsic sounds, and extrinsic sounds was published in that year (Spence, 2012; see Table 1). In addition, the following specific inclusion criteria were instituted during manual examination of titles and abstracts:

- The paper examined the relationship between auditory stimuli (and possibly other sensory modalities) and gustatory and/or olfactory experiences.
- The paper examined the relationship between auditory stimuli and expectations of gustatory features.
- The paper examined the relationship between auditory stimuli and taste terms.

Concerning exclusion criteria, the subsequent were used:

- The paper was purely theoretical (e.g., proposal, review).
- The paper disclosed no experimental studies.
- The paper disclosed no auditory stimuli or no chemosensory attributes in the studies.
- The paper did not assess associations between the auditory and chemical senses.

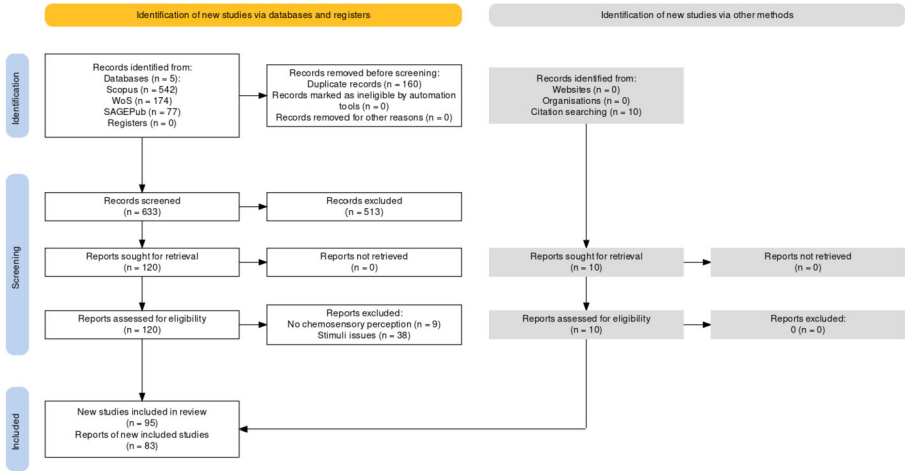


Figure 1. PRISMA flow diagram representing the stages of record identification, screening, and inclusion with number of documents.

- The paper disclosed no chemosensory perception evaluation.
- The paper did not provide information about the stimuli.

Figure 1 portrays a comprehensive diagram of the study selection process. The search yielded a total of 793 records (542 in Scopus, 174 in Web of Science, 77 in SAGEPub, and 0 in MDPI and Emerald), of which 160 were identified as duplicates, resulting in 633 unique records. Titles and abstracts of these records underwent analysis in accordance with the selection criteria. Thus, 513 were omitted due to non-compliance with the criteria, leaving 120 for comprehensive text analysis. Of the original 120 documents, 47 were excluded based on the exclusion criteria, resulting in 73 documents for the data collection process. In addition, 10 documents that did not match the described search parameters were retrieved based on identification across references and review articles, bringing the total to 83 documents for the definitive data processing.

2.4. Structure of the Data Collection Process

In general, Fig. 2 illustrates a schematic representation of the structure of the analysis included in the results section. The experimental methodology of studies was a major category for the analysis of the selected documents. Nonetheless, as this work aims to review the studies focusing on the relationship between the stimuli, two distinct categories were established for the auditory and chemosensory stimuli. The structure was completed by the findings segmented into perception associations and personal preference associations. Perception associations involve sounds principally modulating the intensity of

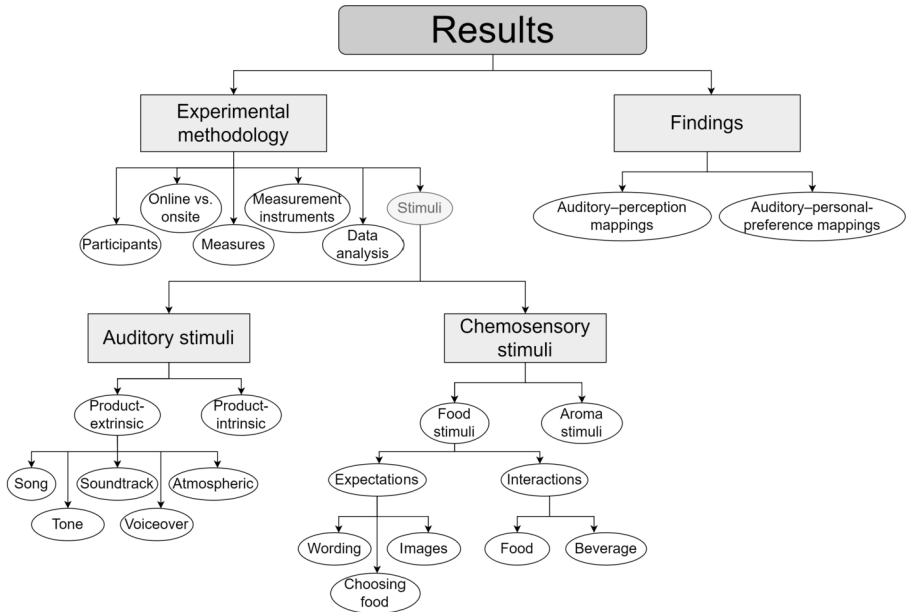


Figure 2. Diagram of the narrative structure of the results section. Square boxes are subsections of results, including auditory stimuli and chemosensory stimuli since the main interest of the work is the interaction between these stimuli.

a flavor or aroma perception. Personal preference associations comprise the modulation of hedonic, emotional, matching, and preference judgments.

In the experimental methodology category, the variables included the participants (number and particular characteristics if explicit), online *vs* onsite setting, experimental design, measurement instrumentation, experimental measures, and data analysis techniques. For the category of auditory stimuli, the analysis was divided by the type of sound into product-extrinsic and product-intrinsic. Product-extrinsic sounds comprise existing commercial songs, custom-made soundtracks (as in sounds composed for experimental purposes), pure or musical tones, atmospheric sounds, and voiceovers. The variables reported for this auditory category were the stimuli description, type of sound, chosen *vs* composed (originally for the corresponding experiment), preliminary intended gustatory correspondence, dominant attributes of the stimuli, and description of panel pretests performed on the stimuli (if applicable).

For the category of chemosensory stimuli, the analysis was divided into food stimuli and aroma stimuli. Aroma stimuli consisted solely of scents, without the inclusion of any food-related wording, imagery, or the actual product. The variables reported here were the stimuli description, dominant attributes of the stimuli, and description of panel pretests performed on the

stimuli (if applicable). Food stimuli were further categorized into expectations (i.e., wording, images, onsite choosing food) and actual experiences with the product (i.e., beverage or food). In addition, they were analyzed through the following variables chosen vs produced, product, and preliminary associations with basic tastes.

Note that if a single publication included multiple experiments correlating auditory stimuli with chemosensory experiences, each experiment was analyzed separately. Note also that several variables were inspired by Spence *et al.* (2019).

3. Results

3.1. Experimental Methodology

On average, the reviewed experiments involved 153 participants, with a standard deviation of 291. In particular, Choi and Cheok (2015) and Mesfin *et al.* (2020) conducted the experiments with the fewest participants involved ($N = 12$), while Wang *et al.* (2021) conducted the one with the most participants involved ($N = 1819$). Some studies involved participants with certain characteristics, such as being experts in a given field (especially wine experts; Wang and Spence, 2018b), trained panelists (Kantono *et al.*, 2016; Lin *et al.*, 2019; Fibrianto *et al.*, 2020), only females (Toffolo *et al.*, 2012; Stafford and Dodd, 2013; Mitchell *et al.*, 2019), only males (Rahne *et al.*, 2018), as well as only elderly population (Endo *et al.*, 2017; Thaploo *et al.*, 2021).

In terms of setting, 89% of experiments were conducted onsite, and 11% were conducted online. Most of the onsite research was conducted in Western Europe, North America, or Oceania. Most experiments (54%) were designed with full within-participants conditions, 30% with both within-participants and between-participants conditions (mixed design), and only 16% with full between-participants conditions.

Concerning measurement instrumentation, the vast majority of experiments (90%) collected data about participants' experiences *via* traditional surveys, either electronic or on paper, and, occasionally, orally. Here, the main instruments used were the Likert scale, the Visual Analogue Scale, multiple choice, and the bipolar scale. On top of that, 10% of the experiments used time-based techniques such as Temporal Dominance of Sensations (TDS; Kantono *et al.*, 2016, 2019; Wang *et al.*, 2019b; Xu *et al.*, 2019; Lin *et al.*, 2022), Temporal Check-All-That-Apply (TCATA; Kantono *et al.*, 2018; Lin *et al.*, 2019; Mitchell *et al.*, 2019), and Time Intensity (TI; Wang *et al.*, 2017b). Some experiments considered a qualitative methodology to measure olfactory perception, using video recordings and interviews as measurement instruments (Choi and Cheok, 2015; Mesfin *et al.*, 2020). Other instruments included

choosing food/beverages onsite, a response button box, an eye tracker, a wristband, a high-performance liquid chromatographic (HPLC) system, and a texture analyzer.

During the experiments, the primary types of measures (dependent variables) were what we term as personal preferences (88%). This category included preference (49%), emotions (especially when aroma stimulus was involved; 40%), matching (32%), familiarity (11%), quality (6%), and engagement (2%), as well as consumer behavior variables (i.e., willingness to pay — WTP, purchase intention, brand recognition; 9%).

Furthermore, 55% of the experiments measured the perceived intensity of the five basic tastes. Sweetness (54%) and bitterness (44%) were measured more frequently than sourness (28%), saltiness (17%), and umami (2%). Beyond taste, multiple flavor-related measures were present in 60% of the experiments, which were gustatory/olfactory intensity, texture, alcohol, fresh, creamy, fruity, temperature, spicy, acidity, temperature, cocoa, milky, body, vanilla, roast, tannins, astringent, crunchy, woody, carbonation, gustatory/olfactory complexity, malty, and richness (see Fig. 3). Finally, as mentioned above, a few experiments included qualitative measures such as participants' facial expressions, tones of voice, gestures, and/or eye gaze.

The analysis of experimental data was conducted using multiple techniques. Eighty-four percent of studies applied analysis of variance (ANOVA), including repeated measures, mixed designs, multivariate, and one-, two-, or three-way models. In addition, *t*-tests, correlations, and chi-squared test techniques were used in 19%, 16%, and 16% of the studies, respectively. Correlations were mainly applied among dependent variables. The remaining analysis techniques were rank-sum, regression, Principal Component Analysis (PCA), Canonical Variates Analysis (CVA), PROCESS macro (of SPSS software),

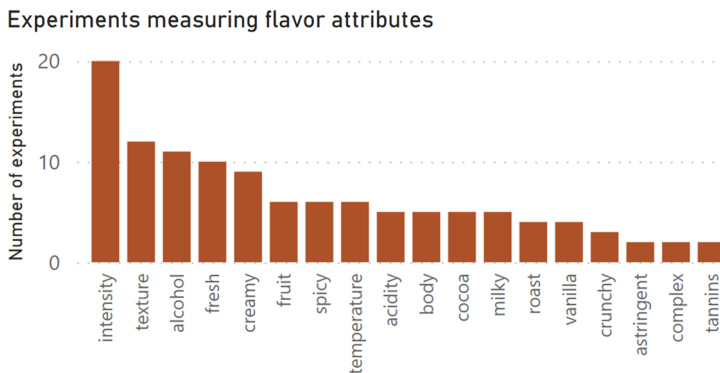


Figure 3. Power BI visualization of number of experiments measuring each principal flavor attribute. Please refer to our Power BI dashboard (<https://bit.ly/visaudchexps>) to interact with this and other multiple visualizations.

Multiple Factor Analysis (MFA), Linear Mixed Model (LMM), Analysis of Covariance (ANCOVA), Bayesian *t*-test, probit model, nonparametric Mann–Whitney *U*-test, log-linear, generalized linear, qualitative, log-likelihood test, Latent Semantic Analysis (LSA), and Correspondence Analysis (CA). Most studies (91%) defined statistical significance at 95 percent confidence. Some studies defined confidence at 99 percent or higher, while a couple of studies defined confidence at 90 percent.

Please consult our interactive Power BI dashboard (<https://bit.ly/visaudchexps>) to explore the visualizations of all data presented in this paper. Additional specific and detailed information about each experiment on these dimensions can be found in Table A1 in the Appendix.

3.2. *Auditory Stimuli*

Product-extrinsic sounds, including commercial songs, soundtracks, pure or musical tones, atmospheric sounds, and voiceovers, were used in almost all experiments (90%). Figure 4 shows the number of experiments that involved each category of product-extrinsic sounds as well as product-intrinsic sounds (refer to Section 1 for a definition of these concepts).

The dominant attributes of the auditory stimuli mainly involved pitch, evoked emotion, timbre, tempo, articulation, consonance, and genre. In approximately one-third of experiments (31%), prior to the main experiment, auditory stimuli were pretested across multiple variables, including gustatory and emotional similarity, as well as genre, tempo, preference, loudness (volume), and country of origin typicality. 41% of the experimental sounds were

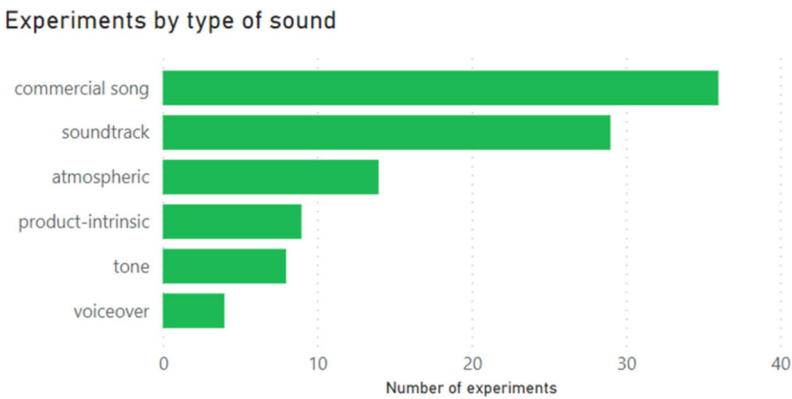


Figure 4. Power BI visualization of number of experiments that used the corresponding type of sound. Please refer to our Power BI dashboard (<https://bit.ly/visaudchexps>) to interact with this and other multiple visualizations.

originally composed for the corresponding study, whereas the rest were pre-existing sounds. It should be noted that some of these pre-existing sounds were digitally manipulated for experimental purposes.

3.2.1. Product-Extrinsic Sounds

Commercial songs were involved in 43% of the experiments focusing on product-extrinsic sounds. They were chosen due to different psychoacoustic features, mainly emotional characteristics, genre, and tempo, as well as pitch, volume, consonance, articulation, timbre, syncopation, roughness, country of origin, distractibility, spectral balance, sharpness, and likability. Furthermore, a few songs were chosen to match specific foods or aromas. For example, Debussy's *Jardin Sous la Pluie* and Rachmaninoff's *Vocalise* were intended to match white wine and red wine, respectively (Wang and Spence, 2015a). In other studies, the songs were chosen to correspond to sour and sweet tastes (Kontukoski *et al.*, 2015; Reinoso-Carvalho *et al.*, 2016; Guedes *et al.*, 2023a). Some of these songs were digitally manipulated specifically for the corresponding experiments (Stafford and Dodd, 2013; Stafford *et al.*, 2013; Fiegel *et al.*, 2014, 2019; Fibrianto *et al.*, 2020; Pantoja and Borges, 2021)

Custom-made soundtracks were used as auditory stimuli in 35% of these experiments. These soundtracks were mainly intended to match taste/flavor attributes, including sweet, bitter, sour, salty, spicy, smooth, and rough (Fig. 5 shows a treemap of the number of experiments considering each taste/flavor). The soundtracks were composed based on features like pitch, consonance, articulation, timbre, tempo, and other psychoacoustic variables. In one study,

Experiments by intended association of sound with flavor

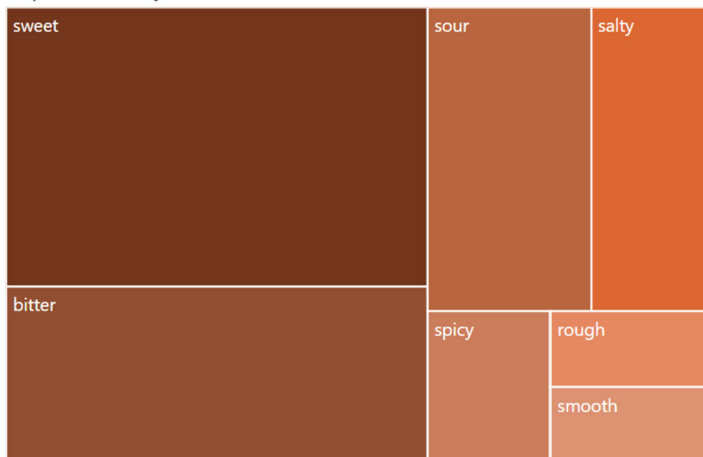


Figure 5. Treemap visualization of soundtrack matches to flavor attributes, sized by experiment count. Please refer to our Power BI dashboard (<https://bit.ly/visaudchexps>) to interact with this and other multiple visualizations.

however, the auditory stimuli were composed to match happy and threatening emotions (Choi and Cheok, 2015). Although soundtracks were mainly composed to correspond to specific sensory or emotional attributes, in some experiments these sounds were exploratory in nature. That is, a variety of sounds were presented to participants to assess their congruence with different flavor attributes (e.g., Knoeferle *et al.*, 2015; Wang and Spence, 2016, 2017, 2018b; Wang *et al.*, 2021).

Atmospheric sounds such as birds, conversations, traffic, construction, machines, leaves rustling, and cabin and white noise, as well as café, countryside, restaurant, bar, park, food court, kitchen, and urban environment sounds were considerably used in different studies (17%). Such sounds were chosen mainly to reproduce urban, rural, typical food consumption, or airplane flight environments, as well as sensory match/mismatch and specific emotions/sensations. In addition, some of them were digitally manipulated in volume and pitch (Ba and Kang, 2019; Bravo-Moncayo *et al.*, 2020; Covaci *et al.*, 2020; Mesfin *et al.*, 2020).

To compare the influence of different pitches on chemosensory experiences, 8% of these studies used tones as auditory stimuli. These experiments were almost evenly split between those involving pure tones and those involving musical tones. Among musical tones, the instrumentation was an important attribute as different instruments were compared, including both Western (i.e., piano, strings, woodwind, and brass; Crisinel and Spence, 2012a, b), and non-Western (i.e., erhu, guzheng, dizi, and yunluo; Qi *et al.*, 2020).

Voiceover was another type of auditory stimulus used throughout these experiments (5%). Researchers used advertisements, spoken news, and nutritional messages to measure their influence on chemosensory experiences. In some of these studies, the recordings were digitally manipulated in pitch or volume (Stafford *et al.*, 2013; Motoki *et al.*, 2019).

3.2.2. *Product-Intrinsic Sounds*

A mere 10% of all experiments used product-intrinsic sounds to measure their influence specifically on food stimulus perception. In most of these studies, sounds of beverage opening, pouring, or effervescence were used as auditory stimuli (Spence and Wang, 2017; Roque *et al.*, 2018; Mitchell *et al.*, 2019; Rodríguez *et al.*, 2021). The sound of mastication was also used to measure perceptions on meals (e.g., crispy/soft, minced-like, fried, simmered foods; Xu and Kerr, 2012; Endo *et al.*, 2016, 2017; Jutras *et al.*, 2019). In one of the latter experiments, the pitch of the actual mastication sound was digitally manipulated (Endo *et al.*, 2016).

Please visit our Power BI dashboard to further engage with numerous visualizations on all data about auditory stimuli (<https://bit.ly/visaudchexps>).

Table A2, in the Appendix, also provides complementary information regarding the auditory stimuli of each experiment across a variety of parameters.

3.3. Chemosensory Stimuli

The chemosensory stimuli used in these experiments were divided into foods (i.e., foods and beverages) and aromas. 87% of these experiments focused on food stimuli, while the remaining (13%) focused on aromas. Figure 6 summarizes the type of chemosensory stimuli used in all experiments. Note that only one study measured the effects of sound on both olfactory and gustatory stimuli, using both sensory cues as independent stimuli (i.e., odorants and tastants; Rahne *et al.*, 2018).

Overall, emotional variables (24%) and alcohol strength (17%) were the most frequently reported attributes of all the chemosensory stimuli. Other dominant attributes of these stimuli are shown in Fig. 7.

3.3.1. Food Stimuli

In the experiments focused on food stimuli, the attributes of these stimuli were mainly their tastes, including sweet (30% of the experiments focused on food), bitter (23%), sour (14%), salty (13%), and umami (3%). Alcohol (11%), emotional attributes (9%), texture attributes (5%), aroma (5%), and spiciness (5%), were also main attributes associated with the food stimuli.

These experiments were almost evenly split between those in which the food stimuli were originally produced for them (41%) and those in which the stimuli were existing foods/beverages (44%). Prior to the main test, 17% of

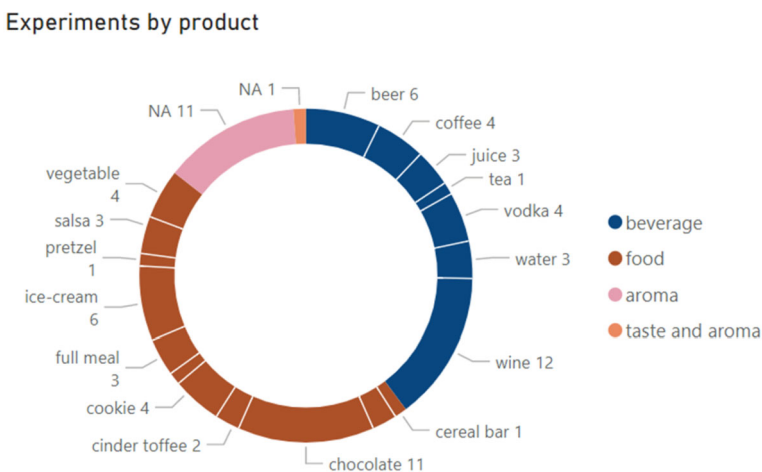


Figure 6. Power BI visualization of number of experiments that used the corresponding product, which was classified between beverage and food. For aromas, the product NA stands for not applicable. Please refer to our Power BI dashboard (<https://bit.ly/visaudchexps>) to interact with this and other multiple visualizations.

Experiments by main attributes of chemosensory stimuli

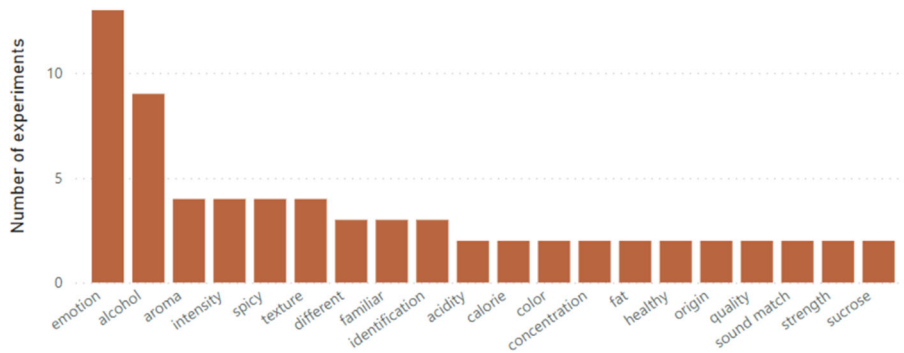


Figure 7. Power BI visualization of number of experiments that reported that their chemosensory stimuli met the corresponding attribute. Please refer to our Power BI dashboard (<https://bit.ly/visaudchexps>) to interact with this and other multiple visualizations.

these studies involved pretesting the corresponding stimuli across multiple variables, including emotional, appropriateness, quality, gustatory, hedonic, mouthfeel, and aroma perception. Most studies (78%) measured actual experiences, while the rest (22%) measured only expectations around the stimuli.

3.3.1.1. Expectations. In 47% of the studies focusing on expectations, flavor stimuli consisted solely of terms, including taste terms (i.e., sour, sweet, bitter, salty, umami; Knoeberle *et al.*, 2015; Wang *et al.*, 2015, 2021; Qi *et al.*, 2020; Rodríguez *et al.*, 2021) and food names (e.g., yogurt, hamburger, sandwich, chips; Motoki *et al.*, 2022). Moreover, one experiment used audio advertisements about cream puff, lemon juice, and black coffee as chemosensory stimuli that attempted to correspond to sweet, sour, and bitter basic tastes, respectively (Motoki *et al.*, 2019). In 17% of these studies, visual cues, namely images of food, beverages, or beverage packages, were the stimuli (Velasco *et al.*, 2014a; Roque *et al.*, 2018; Peng-Li *et al.*, 2020; Padulo *et al.*, 2021; Pantoja and Borges, 2021; Rodríguez *et al.*, 2021).

In 12% of these experiments, expectations were measured by requesting participants to select actual foods and beverages (Chambaron *et al.*, 2015; Kontukoski *et al.*, 2015). One experiment evaluated the choice between high- and low-energy density foods (Chambaron *et al.*, 2015), whereas the other experiment involved the preparation of a preferred juice mixture from sour and sweet juices (Kontukoski *et al.*, 2015).

3.3.1.2. Actual Tasting. Beverages. Most of the studies focusing on actual tasting used beverages as stimuli (52%). These beverages included wine, beer, juice, coffee, vodka, tea, carbonated beverages, and water.

Wine was the beverage most used (38%), including red and white. Wines were mainly selected based on their alcohol, acidity, sucrose, aroma, body, and consumer preference, as well as for being sufficiently different from each other. A few experiments conducted pre-tests on the wines, assessing features like color, aroma, flavor, mouthfeel, and aftertaste perception, as well as hedonistic attitude (Burzynska *et al.*, 2019; Wang *et al.*, 2019a).

Beers were used as beverage stimuli in 19% of these experiments, which contained from 0.5% to 9% of alcohol by volume. Only one beer was produced for the corresponding experiment (Reinoso-Carvalho *et al.*, 2016).

Coffees, including cold and hot brews, were used in 13% of these experiments. Coffee samples were produced considering their sucrose level, aroma, intensity, smokiness, smooth mouthfeel, and/or considering the harvesting, grinding, and roasting conditions.

Vodkas were also used in 13% of these experiments, including cranberry vodka mixes. These vodkas were selected to correspond with atmospheric conditions or to have a sweet–bitter balance. Some of these experiments involved a pilot test to select the vodka brand.

Fruit juices included grape, orange, apple, pear, peach, mango, grapefruit, and pineapple juices (9% of these experiments). Most juices were fruit mixtures produced to be unfamiliar, minimizing the possibility of participants' guessing their nature. Apple and orange were the fruits most used.

Water was used as a beverage stimulus in 9% of these experiments, including distilled water and water solutions. Distilled water was used to pair beverage temperature specifically with music pitch and tempo (Wang and Spence, 2017). Water solutions were, on the one hand, sugar and citric acid water solutions producing a recognizable taste to compare them with wine flavor (Hauck and Hecht, 2019). On the other hand, water solutions were sucrose, NaCl, citric acid, MSG, and quinine-HCl samples of aqueous food grade solutions covering the five tastes, being served at three different levels of concentration (Yan and Dando, 2015).

Green tea with 1, 3, and 5 minutes of brewing times was used to measure gustatory, olfactory, and emotional perceptions in one study (Fibrianto *et al.*, 2020).

Foods. Foods were used as stimuli in 48% of the experiments focused on actual tasting (e.g., chocolate, ice cream, vegetables, cookies, meals, salsa, cinder toffee, and chips). Chocolate was the most popular (32%), including dark, milk, and custom-made varieties. The dominant attributes of chocolate stimuli were evoked emotions, color bias, and tastes (i.e., sweet, bitter, sour, salty).

Chocolate ice creams were used as stimuli in 19% of these experiments, most of which were chocolate gelato. The chocolate flavor was selected

because of its emotional appeal, with bittersweet being the leading gustatory characteristic since it makes it easier to measure the modulatory effect of sound on flavor.

Vegetables such as carrot, tomato, wild rocket, red bell pepper, and cucumber were used as food stimuli in 16% of these experiments. Their main attributes were crispness, softness, and freshness, as well as sugar content and non-emotionality. Pilot tests in these experiments focused on non-emotionality and on perceivable differences in quality (Fiegel *et al.*, 2014; Sinesio *et al.*, 2018).

Cookies were used in 16% of these experiments, including chocolate-chip, vanilla, and soft cookies. Texture, quality, appearance, and sugar content were their main attributes. Pilot tests were conducted to validate whether cookies differed in hedonic response but were at the same time similar in appearance (Hathaway and Simons, 2017).

In 10% of these experiments, stimuli were meals such as nurse care food or a salad-based dish. These meals were mainly nursing-care meals, with an emphasis on texture and a balanced spicy-sweet flavor.

Salsas/dressings were used in 10% of these experiments, including spicy, mild, tomato, and ranch dressing. They were selected for being spicy at different heat levels or easily comparable. Only one salsa mixture was produced for the corresponding experiment (Wang *et al.*, 2017a).

The other foods were chips, cinder toffee, pretzels, cereal bars, and fruits. Chips, pretzels, cereal bars, vegetables, fruits, and cookies were used in one single experiment based on their texture (Jutras *et al.*, 2019). In another experiment, deep-fat fried chips and dried corn chips were used according to their crispness and healthiness (Xu and Kerr, 2012). Finally, two different studies involved home-made cinder toffees (Crisinel *et al.*, 2012; Höchenberger and Ohla, 2019).

3.3.2. *Aroma Stimuli*

As mentioned above, 13% of studies used only aromas. Thirty-six percent of these studies involved pretesting the corresponding aroma prior to the main test across multiple variables, including familiarity, intensity, and emotional perceptions.

The aromas used in these experiments were lemon, raspberry, coffee, lavender, vanilla, almond, caramel, musk, pepper, smoke, tea, garlic, mint, orange, bergamot, lily, apple, apricot, blackberry, cedar, dark chocolate, cut hay, green pepper, honey, licorice, mushroom, pineapple, violet, lilac, osmanthus, bread, cinnamon, onion, menthol, fennel, burnt smell, mildew, cream, grass, menthcamphorate, rodent repellent, sandalwood, watermelon lollipop, soil,

whiskey, roasted sweet potato, mothball, cucumber, leather, green tea, grapefruit, milk, coconut, soap, jasmine, gasoline, valerian root, chamomile, rosemary, peppermint, cassis, and the fragrances Love and Citron. The aromas were released by perfumes, actual objects, sniffing sticks, or aroma diffusers. In addition, some essences were diluted in different solvents before the experiment (Toffolo *et al.*, 2012; Seo *et al.*, 2014; Thaploo *et al.*, 2021). Most studies (73% of these studies) included six or fewer aromas among their olfactory stimuli, while the rest included more than 15 aromas. The stimuli most commonly used were lemon (45%), coffee (36%), raspberry (36%), and lavender (27%).

The aromas were selected based on emotional attributes (i.e., valence and arousal, in 45% of these studies), intensity or concentration (36%), audio or visual congruence (27%), and covering a wide olfactory range (27%). Emotion was measured in most studies (82%), followed by sound–olfactory congruency (64%). In terms of auditory stimuli, 45% of these experiments focused on a comparison between different sound pitches. In some experiments, participants' facial expressions, tones of voice, gestures, and/or eye gaze were measured while they were exposed to aromas (Choi and Cheok, 2015; Mesfin *et al.*, 2020).

Visit our Power BI dashboard (<https://bit.ly/visaudchexps>) to further engage with several visualizations related to the chemosensory stimuli data offered in this section. Tables A3 and A4, in the Appendix, may be of use since they describe information regarding the food and aroma stimuli in each experiment across a variety of variables.

3.4. Findings

As this work aims to review the studies focusing on the relationship between the auditory and chemosensory stimuli, the significant effects reported in crossmodal associations between sounds and chemosensory experiences are reported below. These effects were found in 80% of reviewed experiments. To explore likely patterns and trends, the associations were grouped across three categories: (1) comparable auditory parameters/experiences, namely, sound properties (e.g., high pitch, consonance, legato articulation, piano, and music likability), (2) the reported modulation of taste/flavor attributes, namely, perceptions (e.g., sweet, bitter, fruity, creamy), and (3) liking, valence, matching, or familiarity with an overall product or aroma, namely, personal preferences (in terms of the product or aroma experienced; e.g., juice liking, chocolate matching).

Multiple sound properties were found to be associated with chemosensory experiences. The primary properties identified included high pitch (23% of experiments), low pitch (18%), consonant harmony (15%), high tempo (15%),

legato articulation (14%), piano instrument (11%), and staccato articulation (10%).

When it comes to the chemosensory experiences found to be influenced by sounds, sweet, bitter, salty, and sour tastes were the most common attributes, followed by fruitiness. Sweetness perception was reported to be associated with consonant harmony (in 13% of experiments), followed by legato articulation (12%), high pitch (10%), piano timbre (9%), as well as emotionally positive music, bells timbre (5% each), and music likability (4%). Bitterness was reported to be associated with low pitch (in 9% of experiments), followed by trombone timbre (5%), consonant harmony, and legato articulation (4% each), as well as low tempo, music dislike, and emotionally negative music (3% each). Saltiness was found to be associated with staccato articulation (in 5% of experiments), followed by mid-pitch (3%), as well as dissonant harmony, consonant harmony, minor mode, low pitch, high roughness, and high tempo (2% each). Sourness was discovered to have associations with high pitch (in 5% of experiments), followed by dissonant harmony, high roughness, and staccato articulation (3% each), as well as high sharpness (2%). Finally, fruitiness was associated with cello and piano timbre, legato articulation, and high pitch (in 2% of experiments each).

Concerning associations of aroma stimuli, coffee, for example, was reported to be associated with birds and conversation sounds (Ba and Kang, 2019), as well as with a legato piano melody (Seo *et al.*, 2014). Piano instrumentation was also associated with sweet and fruity aromas, as well as low acidity, spiciness, and bitterness olfactory perception (Crisinel and Spence, 2012a). Birds and conversation sounds were consistently associated with bread and lilac aromas as well (Ba and Kang, 2019). Lemon, raspberry, and menthol aromas were associated with high pitch; onion aroma with low pitch (Speed *et al.*, 2021); osmanthus aroma with bird sounds; and olfactory intensity with urban sounds (Ba and Kang, 2019). Cinnamon and cassis aromas were reported to be associated with specific songs: *Jingle Bells* Christmas Carol (Seo *et al.*, 2014) and *Flowing Streams* by M. Simmonds (Toffolo *et al.*, 2012), respectively. In terms of the personal preferences dimension, olfactory discomfort was reported to be boosted by urban, traffic, and conversation sounds (Ba and Kang, 2019), as well as olfactory unpleasantness by brass instrumentation and pleasantness by piano instrumentation (Crisinel and Spence, 2012a).

To sum up the top findings, Fig. 8 shows the associations between auditory parameters and chemosensory perceptions more frequently confirmed throughout the reviewed experiments. By that, we mean associations that were demonstrated by at least three different experiments.

To interact with various visualizations pertaining to demonstrated cross-modal associations on both perception and personal preference dimensions,

Auditory	Perception	# exps
consonant harmony	sweet	12
legato	sweet	9
low pitch	bitter	8
high pitch	sweet	8
piano	sweet	8
trombone	bitter	5
staccato	salty	5
high pitch	sour	5
bells	sweet	5
positive music	sweet	5
consonant harmony	bitter	4
legato	bitter	4
music likability	sweet	4
low tempo	bitter	3
music dislike	bitter	3
negative music	bitter	3
mid pitch	salty	3
dissonant harmony	sour	3
high roughness	sour	3
staccato	sour	3
distorted timbre	spicy	3
high pitch	spicy	3
high tempo	spicy	3

Figure 8. Power BI table visualization of auditory–perception associations more robustly confirmed, with at least three experiments reporting them. Please refer to our Power BI dashboard (<https://bit.ly/visaudchexps>) to interact with this and other multiple visualizations.

kindly access to our Power BI dashboard (<https://bit.ly/visaudchexps>). In addition, Table A5, in the Appendix, complements information about these associations found in each experiment.

4. Discussion

The present work provides a comprehensive and systematized overview of up-to-date studies focusing on both musical and nonmusical sounds corresponding to chemosensory experiences. Here, 83 documents published between 2012 and 2023, comprising 95 experiments, were characterized across all their methodological attributes, thus identifying the methods and materials currently used to test the influence of sounds on the experience of the chemical senses. In addition, 329 unique auditory–chemosensory associations were identified and analyzed throughout all experiments to explore patterns and trends in these associations, validating the large body of research on cross-modal associations between these sensory modalities.

Offline setting, within-participants experimental design, collecting data through Likert scales in surveys, and analyzing them through ANOVA seem to be the existing methodological standard in this field of research. The measures most used are sweetness, preference, bitterness, and emotional

attributes. When it comes to the auditory stimuli used, commercial songs and custom-made soundtracks are the principal types of sound stimuli. Regarding chemosensory stimuli, most studies focus on food stimuli (i.e., beverages and foods) rather than aromas, with wine and chocolate being the main products used so far. Most studies focused on the actual tasting experience rather than expectations. For both auditory and chemosensory stimuli, an important part of the experiments involves using stimuli originally composed/produced for the experiments, and pilot tests for validation prior to the main experiment.

The reviewed studies provide sufficient information on flavor/aroma perceptions such as sweetness, bitterness, saltiness, sourness, and fruitiness being consistently associated with auditory parameters such as pitch, consonance, articulation, instrumentation, and tempo. For example, sweetness is broadly mapped to consonant harmony, legato articulation, high-pitched sounds, and piano instrumentation. Bitterness is mapped to low pitch, trombone instrument, consonance, and legato articulation. Saltiness is mapped to staccato articulation and mid-low pitch sounds.

As mentioned above, one of the main novelties of this work is that all these results can be explored in various ways through an interactive Power BI dashboard comprising multiple visualizations, including multi-dimensional filters (access <https://bit.ly/visaudchexps> to interact with it).

4.1. Methodological Implications

It is becoming widely recognized that flavor perception is a construction that involves all senses, including audition (Spence, 2015; Spence *et al.*, 2019). There is also enough evidence that olfactory perception depends on the multisensory context and expectation (Calvert *et al.*, 2004; Stein, 2012; Deroy *et al.*, 2013; Olofsson *et al.*, 2019). In fact, the belief that certain fragrances are associated with certain musical tones dates back 150 years (Piesse, 1860). Nevertheless, the crossmodal associations between aromas and sound have received far less attention (Velasco *et al.*, 2014b; McGann, 2017) than flavor-sound experiences. As appreciated, in the present review, only 13% of the existing body of experiments involved aroma stimuli (see Section 3.3.2).

Concerning the characteristics of the participants, the role of individual differences in auditory perception of musical and non-musical sounds has not yet been fully answered (Wang and Spence, 2018a; Spence *et al.*, 2019; Spence, 2022a). Gender, age, cultural background, musical preference, and expertise could indeed mediate music perception (Doty and Cameron, 2009; Eerola and Vuoskoski, 2013; Spence *et al.*, 2019). In the present review, some experiments addressed this issue by selecting or composing sounds based on pleasantness, unpleasantness, likability, dislike, emotionally positive, and emotionally negative perceptions. Moreover, some studies included only participants with particular characteristics (e.g., elderly, male, wine expert,

female, trained), disaggregating the general population into smaller sets to allow specific analyses. Nevertheless, more data are needed to enrich this part of the discussion.

Researchers have reported that auditory and olfactory perceptions may be connected through emotional/hedonic similarity between cues (Crisinel and Spence, 2010; Seo and Hummel, 2011; Deroy *et al.*, 2013). Indeed, 75% of the reviewed studies focusing on aroma stimuli measured emotional perceptions. However, these studies lacked assessing the perceived quality of aromas (cf. Seo and Hummel, 2011).

Relevant discoveries of the present review regarding the methodological attributes might have effects in both scientific and managerial communities. Researchers in this field can find such standard attributes helpful while designing their own experiments. They can either follow them as validated frameworks, or perhaps use them as a starting point toward new related discoveries. For example, researchers might want to compose auditory stimuli to maintain this standard practice and perhaps to go beyond by contributing to new discoveries related to umami perception (a taste scarcely covered by this kind of research; see Sections 3.1 and 3.3). Moreover, marketers and designers are encouraged to rely on the dependability of the methods reported here while developing new multisensory marketing strategies based on auditory cues, with the certainty that multiple findings can be easily replicable. Our interactive Power BI visualizations might encourage such stakeholders to explore different methodologies.

4.2. Theoretical Implications

The scope of the auditory–chemosensory associations reported encompasses a broad range of sounds, foods, and aromas, rendering them applicable across numerous contexts. Auditory stimuli span from commercial songs to ambient noise and chewing sounds. For example, background noise was associated with lilac aroma as well as olfactory discomfort and intensity (Ba and Kang, 2019). While the auditory parameters connected to olfactory experiences have typically been limited to pitch, musical instruments, background music, and white noise (Mahdavi *et al.*, 2020), Ba and Kang (2019) expanded the array of these parameters, incorporating atmospheric urban sounds.

Similarly, food stimuli span from wording and images to actual foods and beverages. In particular, the association between sweetness perception and auditory parameters appears to remain the most widely agreed upon, which is in line with Guedes *et al.* (2022) and Spence *et al.* (2019). While sweetness, bitterness, and sourness are extensively studied attributes, research is also delving into more complex flavor attributes, including alcohol strength, freshness, body, and creaminess. This expanded exploration also encompasses additional food and beverage qualities, including temperature and even wine

and chocolate styles (Spence *et al.* 2019). Despite the limited exploration of umami, a few studies have linked it to the sound of the Chinese instrument *yunluo* (a set of small tuned gongs mounted in a wooden frame) and airplane cabin noise (Yan and Dando, 2015; Qi *et al.*, 2020).

Aromas also span a wide variety of olfactory notes. For example, Speed *et al.* (2021) reported that raspberry, menthol, and lemon aromas are associated with higher pitch tones than caramel and onion aromas. This aligns with the fact that aromas being easier to identify tend to be consciously associated with high-pitched sounds (Stevenson *et al.*, 2012).

The associations reported in the present review align with potential mechanisms through which sonic seasoning soundtracks can give rise to perceptual differences (Wang, 2017); for example, sensory expectations (Velasco *et al.*, 2014a; Padulo *et al.*, 2021; Rodríguez *et al.*, 2021), attention capture (Wang *et al.*, 2017b, 2019b), and emotion mediation (Reinoso-Carvalho *et al.*, 2016, 2019, 2020a; Vandenberghe-Descamps *et al.*, 2022). Furthermore, these experiments also address potential explanations for the origin of auditory–flavor associations as proposed by Knöferle and Spence (2012). For instance, intensity matching across sensory modalities (Fiegel *et al.*, 2014, 2019; Wang *et al.*, 2019b) and semantic matching (Crisinel and Spence, 2012a; Chambaron *et al.*, 2015; Kontukoski *et al.*, 2015; Vandenberghe-Descamps *et al.*, 2022).

In certain studies, companies collaborated or provided funding. However, their involvement was reported and carefully controlled, ensuring ethical behavior, with all research methodologies adhering to scientific standards.

4.2.1. *Likely Inconsistencies of Auditory–Chemosensory Associations*

While the current body of research presents numerous relevant and consistent findings, it also showcases significant inconsistencies concerning the crossmodal associations. In particular, some experiments exhibited a variable association between the sweet taste and sound parameters (such as pitch or sharpness), depending on the comparative taste (Velasco *et al.*, 2014a; Kontukoski *et al.*, 2015). The combination of different sound parameters within a single sound can also drive some inconsistencies between low-level parameters. For example, the influence of pitch appears to be more significant in shaping sweetness perception than volume, resulting in both low and high volume being associated with sweetness (Stafford *et al.*, 2012, 2013; Rodríguez *et al.*, 2021). Such inconsistency was also apparent with the perceptions of bitterness, saltiness, sourness, acidity, and wine liking (Wang and Spence, 2015a, 2018b; Wang *et al.*, 2015, 2020).

Other potential factors of exhibited inconsistencies are the influence of the absence of sounds, the specific type of sound, or the category being measured. For instance, perceptions of bitterness, sourness, and astringency were associated with opposing sound features, as an experiment found that the presence

of music (vs its absence) evoked these perceptions (Fibrianto *et al.*, 2020). Furthermore, the impact of sound on sourness perception might vary when the sound is a voiceover (Motoki *et al.*, 2019). Fruitiness perception shows opposing associations with various musical parameters, likely due to the diversity of products examined, such as wines and raspberry/citron vodkas (Wang and Spence, 2015a, b; Wang *et al.*, 2019a). While fruitiness has been evaluated in multiple instances during the tasting/smelling of alcohol drinks (e.g., wine, vodka, beer), its interpretation might be confounding due to the diverse tastes that fruits can encompass, such as acidic and sweet. Overall, the reported inconsistencies can be attributed to the relativity of the demonstrated effects, as well as the consideration of the co-occurrence of auditory attributes within the same sound.

4.3. Limitations and Research Opportunities

4.3.1. Opportunities in the Experimental Design

The primary concern with respect to chemosensory experiences is the notable scarcity of studies focusing on assessing standalone olfactory experiences as well as overall expectations. Measuring these experiences would be quite useful for food and other industries (e.g., cosmetics). In addition, most experiments have focused on actual experiences rather than expectations (see Section 3.3), mostly underestimating the moment of product choice, in which expectations become of importance and the integration of sounds and aromas can have a crucial role since taste is not involved (Wang and Spence, 2018a). Therefore, expectations should be further assessed in experiments associating sounds with chemosensory experiences.

Concerning the population, only three experiments in the present review involved non-Western (e.g., Asian) and Western participants combined (Knoeflerle *et al.*, 2015; Peng-Li *et al.*, 2020; Reinoso-Carvalho *et al.*, 2020a). The scarcity of cross-cultural assessments is to be expected, considering how difficult such initiatives are. However, one might expect that, thanks to increasing globalization and digital instrumentation, cross-cultural approaches will become more usual, for instance, *via* online studies.

In 2010, Henrich *et al.* (2010) drew the attention to the fact that samples in experimental psychology research were drawn entirely from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) societies although they are among the least representative populations one could find while generalizing human behaviors. Since most of the onsite research was conducted in Western Europe, North America, or Oceania, disaggregating the general population into smaller sets can be helpful to further explore the behavior of populations that have been underrated as well, such as Asians, Africans, Latin Americans, elders, children, or people with a low educational level. Expanding

research on these populations is crucial as it is mostly unclear which auditory–chemosensory associations are culturally dependent (Qi *et al.*, 2020; Guedes *et al.*, 2023b).

Working with specific subsets of the population may also be useful due to the distinct needs that each population may demand. Older people with partial or total loss of olfaction can benefit from auditory–chemosensory crossmodal associations (Deroy *et al.*, 2013), say, using sounds as substitutes for smell, with the sound intensity varying depending on the sniffing rate and specific timbres corresponding to the pleasantness of scents (Cooke and Myin, 2011; Mahdavi *et al.*, 2020).

In the context of music, classical music appears to be the prevailing genre typically used in music-and-emotion-oriented and mood-induction studies (Eerola and Vuoskoski, 2013). Although the present review does not show an evident predominance of a specific musical genre across the commercial songs, Western genres were used in almost all studies. On the other hand, Eastern genres were only explored in one experiment in this review (Zhang *et al.*, 2016), and another explored Chinese instruments (erhu, guzheng, dizi, and yunluo; Qi *et al.*, 2020).

Regarding measurement instrumentation, most data came from self-reported measurements. Some of these measurements (11%) were time-based, including Temporal Dominance of Sensations (TDS), Temporal Check-All-That-Apply (TCATA), and Time Intensity (TI). These time-based techniques allow for examining the changes in the dominance of a pre-defined list of sensory attributes across time. Hence, greater involvement of such measures may be crucial due to the likely dynamism of these senses when exposed to a continuous stimulus. However, researchers should be aware that such measures require beforehand time for panel familiarization with such methods. On top of that, sensory perception processes can also be implicit and unconscious. Thus, recording and analyzing reaction times would also be helpful for assessing implicit consumer responses.

4.3.2. *Challenges in Selecting and Reporting Auditory Stimuli*

Further exploring individual perceptions related to music, such as music likability, music dislike, emotionally positive music, and emotionally negative music, may be relevant for conducting individualized interventions (Guedes *et al.*, 2023b). In fact, such subjective emotional attributes have been proven to mediate the relationship between sounds and chemosensory experiences through the process of sensation transfer or affective priming (Spence *et al.*, 2019; Guedes *et al.*, 2022). This can be approached by building personal lists based on each participant's perceptions or even by processing personal playlists on digital platforms like Spotify, Last.fm, or Xiami Music.

To overcome the reported inconsistencies in terms of the auditory-gustatory crossmodal associations in the future (see Section 4.2.1), researchers might consider including absolute values of low-level auditory parameters (e.g., pitch, loudness, discontinuity, tempo, instrumentation) rather than relative values. Although the evidence supports that crossmodal correspondences involving the auditory dimension of pitch are mainly relative (Spence, 2019b), reporting the absolute values of the stimuli used in the experiments might facilitate comparisons across experiments. Researchers could, for example, describe the stimuli in terms of a measure (e.g., Hz, dB, bpm; Endo *et al.*, 2016; Wang and Spence, 2017; Motoki *et al.*, 2019; Covaci *et al.*, 2020; Pantoja and Borges, 2021), rather than describing them relative to other stimuli (high *vs* low). Moreover, disaggregating features like consonance or articulations into parameters of lower level (e.g., roughness, length of attack) would be helpful in achieving this objective (Reinoso-Carvalho *et al.*, 2015b).

The rationale given by the authors behind selecting or composing/producing the stimuli was essential for making the comparative analysis in the present work. Therefore, future studies would help such comparative works by explicitly justifying their choices or thoroughly describing the dominant attributes of stimuli.

4.3.3. Opportunities With the Dataset of Crossmodal Associations

One of the main outcomes of the present review is going beyond the usual auditory-flavor associations by systematizing a large dataset of 329 auditory-chemosensory associations that include both perceptual and personal preference records. Refer to Fig. 8 for the top associations, which can be further explored and filtered in our Power BI dashboard (<https://bit.ly/visaudchexps>), as well as Table A5, in the Appendix, for all the associations found in each experiment. With such free, open-access data, the auditory parameters reported can be used to rethink soundscapes, as well as for the design of sounds generated by the interaction with the packaging or the product, making them useful for both choice and consumption/usage contexts. With this important resource, researchers will be able to take more reliable and purposeful steps toward understanding consumer chemosensory experiences, not only for food or beverages but also for other industry sectors, such as cosmetics.

As there can sometimes be a commercial pressure to bypass the pursuit of scientific validation (Spence *et al.*, 2021), Fig. 8 provides quick and quantitative robust scientific foundations of principal crossmodal associations for product-intrinsic/extrinsic sound design applications. Hence, these auditory-chemosensory associations, along with the interactive visualizations in the Power BI dashboard, may have compelling practical implications. For example, advertisers can produce songs, pure/musical tones, atmospheric sounds, voice recordings, or product-intrinsic sounds that can properly convey food,

beverage, perfume, scented candle, and cosmetics characteristics, thus augmenting specific product attributes through the sense of hearing.

However, it is important to recognize that when science is misinterpreted, the resulting sound compositions and choices can easily end up being haphazard, underwhelming, and perhaps even confusing to the target (Spence *et al.*, 2021). Such a result may not only be counterproductive for brands but also for the consumer perception of crossmodal science (Spence *et al.*, 2021). In fact, the present work also draws attention to some inconsistencies in terms of the crossmodal associations (see Section 4.2.1). As a result, companies should be careful with the associations provided and consider the comparisons for which the associations were significant in the corresponding experiment and the co-occurrences of the auditory attributes (see Table A5).

As future work, there is still an opportunity to conduct a meta-analysis, pool the results of this evidence, and estimate a common effect size through additional statistical techniques. Like this, it may be possible to draw more precise conclusions about the effects of each of the summarized associations. Thus, the present work serves as the raw material for researchers interested in such future outcomes.

4.3.4. *Opportunities for Technological Application*

Through its methodological and comparative focus, the present review uncovers a conspicuous lack of application of computational models in guiding the experimental methodology design of this research. In fact, 84% of experiments analyzed their data with ANOVA techniques (see Section 3.1), and their auditory stimuli selection was primarily based on researchers' expertise, pilot tests, and previously published studies (see Section 3.2). Nonetheless, a recent study by Kantono *et al.* (2022) introduced a novel proof-of-concept utilizing neural networks for data analysis in an experiment. This experiment explored varying degrees of illumination and of background sound level within a virtual-reality restaurant setup while assessing situational appropriateness and food wanting. The authors then used the particle swarm optimization algorithm to identify the optimal restaurant lighting and sound level for consuming burgers and pizzas.

The latter reveals an outstanding opportunity to leverage cutting-edge technologies for the quantitative nature of this review findings, and mainly the open-access dataset of crossmodal associations. These technologies could lead to the refinement of the methodological process and encourage the involvement of companies in the scientific validation of their communication strategies through sound. For example, think of a computational model approach to parameterize the relationship between sounds and chemosensory experiences, enhancing the accuracy of the interventions. Another example is a learning

generative model to produce or predict sounds with specific multisensory perceptual targets. As the amount of data grows, these modeling techniques may more accurately represent and predict such relationships.

Curated playlists with the necessary sonic properties to correspond to chemosensory experiences are currently offered *via* online music platforms such as Deezer, Spotify, Pandora, and Google Play Music (Spence *et al.*, 2021). Given the growing literature, there is a growing interest in using technology to synchronize aspects of auditory stimulation for augmented multisensory tasting. Practical examples of potential applications have already been successfully brought to the market, such as The Sound of Chocolate in Belgium, where chocolate boxes were sold alongside bespoke soundtracks; sensory apps connecting products with specific pieces of music; and technologically enhanced cups attempting to reduce sugar intake *via* sound (Spence *et al.*, 2019).

Such technological approaches can go far beyond the food industry. In fact, it seems that the cosmetic industry may be the next industry where this type of knowledge may bloom (see Spence, 2021, 2022b for studies in perfume-music matching). So far, sonic marketing activations for cosmetics have been lacking in scientific validation (Spence, 2021). Notably, such activations may operate in both directions — the addition of scent can modify people's responses to music, while what people hear can also influence what they think about what they smell.

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Author Contributions

Brayan Rodriguez: conceptualization, methodology, investigation, data curation, writing — original draft, visualization. Felipe Reinoso-Carvalho: conceptualization, validation, resources, writing — review and editing, visualization, supervision. Luis H. Reyes: conceptualization, validation, writing — review and editing, visualization, supervision.

Declaration of Competing Interest

The authors declare that they are not aware of competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Notes

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Appendix

Table A1.

Methodology dimensions of experiments excluding stimuli. N: N = number of participants. L/S: L = online, S = onsite. Design: W = within participants, B = between participants, M = mixed model. Measurement instrumentation: TDS = temporal dominance of sensations, TCATA = temporal check-all-that-apply, TI = time intensity. Measures: WTP = willingness to pay, SAM = Self-Assessment Manikin, PANAS = Positive and Negative Affect Schedule (interested, excited, strong, enthusiastic, proud, alert, inspired, determined, attentive, active, distressed, upset, guilty, scared, hostile, irritable, ashamed, nervous, jittery, afraid)

Reference	N	L/S	Design	Measurement instrumentation	Measures	Data analysis techniques
Crisinel and Spence (2012a)	30	S	W	survey	Match pitch–aroma, match instrument–aroma, describing aroma (complex, familiar, intense, pleasant, acrid, earthy, floral, fruity, nutty, spicy, woody, bitter, salty, sour, sweet), aroma identification	Mixed-model ANOVA, PCA, chi-square test, one-way ANOVA
Crisinel and Spence (2012b)	44	S	W	survey	Sound matched, bitterness, sweetness, pleasantness	Repeated-measures ANOVA, chi-square test
	38	S	W	survey	Sharp–blunt, light–dark, rough–smooth, unpleasant–pleasant, sour–bitter, simple–complex, crunchy–creamy, clean–dirty, fresh–stale, sweet–salty, unfamiliar–familiar, best adjective match with high/low pitch, ratings match with pitch	Repeated-measures ANOVA, chi-square test
Crisinel <i>et al.</i> (2012)	20	S	W	survey	Bitter–sweet, taste likability, place of flavor in mouth	Mean and standard deviation
North (2012)	250	S	B	survey	Powerful/heavy, subtle/refined, zingy/refreshing, mellow/soft, wine liking, match music	Multivariate ANOVA
Stafford <i>et al.</i> (2012)	80	S	M	survey	PANAS, cold, familiar, alcohol strength, like, sweet, bitter	Repeated-measures ANOVA

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Toffolo <i>et al.</i> (2012)	70 ¹	S	M	survey	Emotional feeling, movie vividness, film pleasantness, film arouse, fresh–stale, good–bad, happy–sad, harmonious–inharmonious, healthy–unhealthy, beautiful–ugly, smooth–rough, clean–dirty, safe–dangerous, strong–weak, powerful–powerless, harsh–mild, ordered–chaotic, quiet–noisy, relaxing–stimulating, wet–dry, muddy–clear	Dependent samples <i>t</i> -tests, one-way between subjects ANOVA
Xu and Kerr (2012)	100	S	W	survey, texture analyzer	Color, texture, flavor, overall likability, color measurement, sound energy/frequency spectrum of fracture sound	ANOVA among groups
Stafford and Dodd (2013)	45 ¹	S	B	clock, survey	Consumption duration, alert, thirsty, drowsy, hungry feelings, PANAS, coldness, familiarity, alcohol strength, likability, sweetness, bitterness	Univariate ANOVA, repeated-measures ANOVA
Stafford <i>et al.</i> (2013)	54	S	M	survey	Alert, thirsty, drowsy, hungry, PANAS, coldness, familiarity, alcohol strength, likability, sweetness, bitterness	Univariate ANOVA, repeated-measures ANOVA, correlations
Fiegel <i>et al.</i> (2014)	99	S	M	survey	Hunger, pre-pleasantness, flavor intensity, pleasantness, texture liking, overall liking, music familiarity, pleasant, arousal	Nonparametric Friedman test, Wilcoxon signed-rank test, nonparametric Spearman correlation, nonparametric Mann–Whitney <i>U</i> -test
Seo <i>et al.</i> (2014)	26	S	W	survey	Aroma intensity, aroma pleasantness, sound–aroma congruency, aroma pleasantness	Pearson correlation, paired <i>t</i> -tests, two-way repeated- measures ANOVA

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Velasco <i>et al.</i> (2014b)	34	S	W	survey, food choice	Reaction time, packaging sweet–sour, sweetness, sourness	Log-likelihood test (G2) of independence with Williams’ correction represented <i>via</i> mosaic plots, repeated-measures ANOVA
Chambaron <i>et al.</i> (2015)	147	S	B	food choice	Food choice for lunch	Generalized linear model with a binomial distribution and logit link with two two-level factors
Choi and Cheok (2015)	12	S	B	video, interview	Subconscious changes of facial expressions, tone of voices and gestures. Impressions and feelings about their communication pattern.	Qualitative analysis
Knoeferle <i>et al.</i> (2015)	39	L	W	survey	Sweet, sour, salty, and bitter match	Repeated-measures ANOVA
	61	L	W	survey	Sweet, sour, salty, and bitter match	Repeated-measures ANOVA
	510	L	W	survey	Sweet, sour, salty, and bitter match	Generalized linear mixed-effects model (GLMM) with binomial error distribution and a log-link function
Kontukoski <i>et al.</i> (2015)	66	S	M	chemical software, survey, food choice	Food-related ideas to music, taste associations (banana–lemon, chocolate–lingonberry, meringue–crisp rye bread, pineapple–red currant), sugar content and organic acid content of beverage preparation congruent with music	Latent Semantic Analysis, one-way ANOVA

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Reinoso- Carvalho <i>et al.</i> (2015a)	299	S	B	survey	Chocolate liking, WTP, sweetness, bitterness, sourness, saltiness	Paired samples <i>t</i> -test
Wang and Spence (2015a)	64	S	M	survey	Wine–music match, fruitiness, acidity, tannins, richness, complexity, length of flavor in mouth, wine pleasantness	Independent samples <i>t</i> -tests, mixed measures ANOVA, Pearson correlations, two-way mixed measures ANOVA
Wang and Spence (2015b)	67	S	M	survey	Vodka–environment match, round–angular shape, vodka liking, fresh–fruity	Pearson’s correlation, repeated-measures multivariate ANOVA, two-way mixed measures multivariate ANOVA, one-way multivariate ANOVA
Wang <i>et al.</i> (2015)	100	L	W	survey	Best match (sweet, sour, bitter, or salty)	Chi-square test, repeated-measures ANOVA
Yan and Dando (2015)	48	S	W	survey	Taste intensity	Student’s <i>t</i> -tests, linear mixed models
Endo <i>et al.</i> (2016)	30	S	W	survey	Light–heavy, cheap–expensive, insipid–rich, stale–fresh, simple–complex, un–palatable, soft–hard, dry–moist, common–rare, not–chewy, boring–interesting, smooth–rough, bad–good texture, un–exciting, un–satisfied, diminishes–arouses appetite, less–more involved dining experience, un–pleasant, sound–food naturalness, comfortable sound, number of ingredients, chewable	Two-way repeated-measures ANOVA

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Kantono <i>et al.</i> (2016)	45 ³	S	W	TDS, survey	Music liking, TDS (amusement, happiness, enjoyment, anger, contempt, disgust, love, surprise, satisfaction), TDS (sweetness, bitterness, saltiness, sourness, umami), gelati pleasantness	Full facotrial ANOVA, chi-square test, Canonical Variate Analyses
Reinoso- Carvalho <i>et al.</i> (2016)	231	S	B	survey	Beer liking, WTP, sweetness, bitterness, sourness, alcohol strength, music–beer match	Mixed ANOVA
Wang and Spence (2016)	39	S	W	survey	Music liking, drink liking, sour–sweet drink taste	Two-way repeated measures ANOVA
Zhang <i>et al.</i> (2016)	18	S	W	survey	Pleasantness of music alone, pleasantness of music with aroma, music–aroma congruency	ANOVA, Pearson correlation analysis
Endo <i>et al.</i> (2017)	30 ⁱ	S	W	survey	Light–heavy, cheap–expensive, insipid–rich, stale–fresh, simple–complex, un–palatable, soft–hard, dry–moist, common–rare, not–chewy, boring–interesting, smooth–rough, bad–good texture, un–exciting, un–satisfied, diminishes–arouses appetite, less–more involved dining experience, un–pleasant, sound–food naturalness, comfortable sound, number of ingredients, chewable, mastication intensity, mastication rhythm	Three-way repeated-measures ANOVA, Wilcoxon signed-rank test
Guetta and Loui (2017)	50	S	W	survey	Music clip–chocolate taste match	Chi-square test

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Hathaway and Simons (2017)	59	S	W	survey	Usability, environmental aesthetics, novelty, involvement, sensory awareness, immersion, distraction, realism, liking, preference	Three-way repeated-measures ANOVA, two-way repeated-measures ANOVA, Friedman’s test with Wilcoxon signed-ranks test.
Reinoso- Carvalho <i>et al.</i> (2017)	116	S	M	survey	Likability, sweet, bitter, creamy	Repeated-measures multivariate ANOVA
Spence and Wang (2017)	140	S	W	survey	Intensity, quality, celebration appropriateness, celebratory mood, cork/screw cap preference	Chi-square test, Pearson’s correlations, multivariate ANOVA
Wang and Spence (2017)	24	S	W	survey	Beverage temperature, pitch/tempo match	Repeated-measures ANOVA
Wang <i>et al.</i> (2017a)	180	S	B	survey	Expectations listening and looking (sweet, spicy, flavor intense, liking), 5 mins after eating (sweet, spicy, flavor intense, liking)	Multivariate ANOVA, Pearson correlations
	18	S	W	survey	10 s after tasting (spicy, flavor intense, pleasantness)	Repeated-measured ANOVA
	40	S	W	survey	Spicy, flavor intense, liking food/music	Repeated-measures analysis-of-covariant, univariate ANOVA
Wang <i>et al.</i> (2017b)	21	S	W	TI	TI (sweetness, sourness)	Area under the curve of TI graphs, repeated-measures ANOVA
Kantono <i>et al.</i> (2018)	253	S	M	TCATA, survey	TCATA (sweetness, bitterness, milkiness, creaminess, cocoaness), amusement, happiness, enjoyment, anger, contempt, disgust, love, surprise, satisfaction while listening to music	One-way repeated-measures ANOVA, multiple factor analysis, two-sided Fisher–Irwin test, Correspondence Analysis, chi-square test

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Rahne <i>et al.</i> (2018)	16 ⁱⁱ	S	W	survey	Scent discrimination, identification and thresholds, taste identification and thresholds	Repeated-measures ANOVA
Roque <i>et al.</i> (2018)	84	S	M	survey	Fresh–no fresh, response time	Linear mixed-effects model
Sinesio <i>et al.</i> (2018)	48	S	W	survey	Overall liking, perceived freshness, 21-item engagement of the whole immersive setting (usability, environmental aesthetic, novelty, involvement, sensory awareness, immersion, realism, distraction)	Three-way ANOVA
Wang and Spence (2018b)	154 ⁱⁱⁱ	S	M	survey	Three top flavors, sweetness, acidity, liking	Average values of ratings, multivariate ANOVA
		S	M	survey	Body, balance, length, wine liking, music–wine match, music liking	Multivariate ANOVA, SPSS PROCESS macro
Wang and Spence (2018c)	49	S	W	survey	Sour–sweet, pleasantness	Repeated-measures multivariate ANOVA, linear regression
Ziv (2018)	60	S	M	survey, food choice	Tasty, sweet, quality, prestigious, refined, pleasant, soft, cookie preferred	Mixed-design ANOVA, orthogonal paired-sample <i>t</i> -tests, chi-square tests
	121	S	M		Tasty, sweet, quality, prestigious, refined, pleasant, soft, cookie preferred	Repeated-measures ANOVA, paired-sample <i>t</i> -tests
Ba and Kang (2019)	168	S	M	survey	Acoustic comfort, sound (preference, familiarity, subjective loudness), aroma (comfort, preference, familiarity, subjective intensity), overall comfort, sound–aroma congruency	Mean and standard deviation, mixed design ANOVA, one-way ANOVA
Burzynska <i>et al.</i> (2019)	50	S	W	survey	Body, acidity, aromatic intensity, liking	Pearson correlation, repeated-measures ANOVA

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Cho <i>et al.</i> (2019)	126	S	B	survey	Overall liking, perceived quality, hunger, fullness, desire to eat, cookie liking, desire to eat cookie, craving, seven items of food involvement	Between-subjects analysis of covariance
Fiegel <i>et al.</i> (2019)	61	S	W	survey	Flavor intense, flavor pleasant, texture liking, overall liking, music pleasant, music stimulation	Nonparametric Friedman test (Kendall's coefficient), Wilcoxon signed-rank test for pairwise comparison, Friedman test, Spearman partial correlation analysis with covariance
Hauck and Hecht (2019)	115	S	W	survey	All for music and beverage separately: global liking, sweet, sour, salty, bitter, foul, floral, aromatic, fruity, lively, gloomy, harmonic, light, zingy/refreshing, powerful/heavy, subtle/refined, mellow/soft	Repeated-measures ANOVA
Höchen-berger and Ohla (2019)	40	S	W	button, survey	Taste intense, pleasant, bitter–sweet, sweet, bitter, salty, sour, yes–no remember, pleasantness recall, bitter–sweet–sour–salty sound association	Repeated-measures ANOVA (with Bayes factors), student's <i>t</i> -test, Bayesian <i>t</i> -test
Hsu and Chen (2019)	40	S	B	food choice	Brainwave data collected by EEG, wine selection	Descriptive statistics, chi square test, paired sample <i>t</i> -test, Pearson's correlation
Jutras <i>et al.</i> (2019)	59	S	W	survey	Freshness, current liking, general liking, further liking	Three-way repeated-measures ANOVA, nonparametric Friedman's ANOVA

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Kantono <i>et al.</i> (2019)	90	S	W	TDS	TDS (sweet–bitter–cocoa–milky–creamy–vanilla–roasted), TDS (amusement, happiness, enjoyment, anger, contempt, disgust, love, disappointment, satisfaction), with electrodes (skin conductance, respiratory rate, blood volume pulse, heart rate)	Partial least square path modelling, dominance rating, Canonical Variate Analysis, multivariate ANOVA, one-way repeated-measures ANOVA, partial least square regression
Lin <i>et al.</i> (2019)	58 ³	S	W	TCATA, survey	TCATA (sweet, bitter, cocoa, milky, creamy, vanilla, roasty), SAM (valence, arousal, dominance)	Fisher–Irwin test, correspondence analysis, chi-square tests of independence, one-way ANOVA, Multiple Factor Analysis
Liu <i>et al.</i> (2019)	50	S	W	survey	Coffee preference ranking, coffee acceptability, auditory recall	Binomial analysis, two-way repeated-measures ANOVA, three-way repeated-measures ANOVA, binomial statistics
Luca <i>et al.</i> (2019)	30	S	M	survey	Sparkling, refreshing, delicate, refined, sweet, sour, alcoholic, pleasant	ANOVA
Mathiesen <i>et al.</i> (2019)	32	S	W	survey	Sweetness, bitterness, liking, sound liking	Repeated-measures multivariate ANOVA
Mitchell <i>et al.</i> (2019)	117 ^{iv}	S	B	TCATA	TCATA (astringent, bitter, carbonation, fruity/hops, malty, sour, sweet)	Area Under the Curve of TCATA plot divided in early, middle and late time points.
Motoki <i>et al.</i> (2019)	59	S	M	survey	Sweetness, sourness, bitterness, sound (liking, pitch perception)	<i>t</i> -test, ANOVA

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Peters Rit <i>et al.</i> (2019)	52	S	W	survey	Brand recognition, excitation (advertisement, brand, product), pleasantness (advertisement, brand, product, aroma, sound)	Separate linear mixed effects models
Reinoso- Carvalho <i>et al.</i> (2019)	221	S	M	survey	Beer liking, sweet, bitter, sour, alcohol, body, WTP	Repeated-measures ANOVA, independent <i>t</i> -test
	154	S	M	survey	Beer liking, sweet, bitter, sour, alcohol, body, WTP	Repeated-measures ANOVA, independent <i>t</i> -test
	157	S	M	survey	Beer liking, sweet, bitter, sour, alcohol, body, WTP	Repeated-measures ANOVA, independent <i>t</i> -test
Wang <i>et al.</i> (2019a)	46	S	W	survey	Fruitiness, spices, smoothness	Repeated-measures ANOVA
Wang <i>et al.</i> (2019b)	21	S	W	TDS	TDS (red fruit, tannins, alcohol, woody, sweet, acidic, spicy, bitter)	TDS curves, pairwise Pearson correlations, nonparametric rank-sum
Wang <i>et al.</i> (2019c)	331	S	M	survey	Beverage liking, sweet, sour, bitter, intensity perception	Multivariate ANOVA, independent samples <i>t</i> -test
Xu <i>et al.</i> (2019)	45	S	W	TDS, survey	Amusement, enjoyment, happiness, love, satisfaction, anger, contempt, disappointment, disgust, electrode (skin conductance, respiratory rate, blood volume pulse, heart rate), TDS (cocoa, milky, creamy, vanilla, roasted, sweet, bitter)	Repeated-measures ANOVA, Canonical Variate Analysis, Hotelling–Lawley multivariate ANOVA, PCA, one-way repeated-measures ANOVA
Bravo- Moncayo <i>et al.</i> (2020)	384	S	M	survey	Coffee liking, sound liking, sound–coffee match, sweetness, bitterness, acidity, flavor intensity, aroma intensity, temperature, purchase intention, WTP	Mixed-design multivariate ANOVA

Table A1.
(Continued)

Reference	N	L/S	Design	Measurement instrumentation	Measures	Data analysis techniques
Covaci <i>et al.</i> (2020)	24	S	M	survey	Smell (relevance, strength, distracting, video consistency, annoyance, faded, enhancing video, overall enjoyment), eye gaze data, heart rate	Independent samples <i>t</i> -test
Fibrianto <i>et al.</i> (2020)	14 ^v	S	W	survey	Emotional (ND), bitter, sweet, sour, astringency, toothetch, jasmine, rose, green herb, smoky, fresh, floral, tobacco, brown, spinach, dried straw	ANOVA
Mesfin <i>et al.</i> (2020)	12	S	B	eye-tracker, video, wrist band, survey	Visual attention (gaze points), heart rate, smell (relevance, strength, distracting, match, annoying, faded away slowly), sound (match, loudness, distracting, annoying, reality enhancing, watching enhancing), watching enjoyment, haptic (relevance, distracting, annoying, reality enhancing, watching enhancing), overall enjoyment.	ANOVA
Nivedhan <i>et al.</i> (2020)	50	S	W	survey	Liking, bitter, sweet	Repeated-measures ANOVA
Peng-Li <i>et al.</i> (2020)	72	S	W	survey, eye-tracker	Eye movement, liking, familiarity, sweet-salty sound match	Hierarchical Linear Mixed Models, chi-square goodness/independence tests, one-way multivariate ANOVA
Qi <i>et al.</i> (2020)	80	S	W	survey	Color/taste term-sound match	Repeated-measures ANOVA, log-linear analysis, one-sample <i>t</i> -tests, correlation, one-way repeated-measures ANOVA, log-linear analysis

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Reinoso- Carvalho <i>et al.</i> (2020a)	1390	S	M	survey	Flavor liking, flavor–music match, sweet, bitter, music liking, flavor strength, purchase intention, chocolate texture, WTP	Multivariate multinomial ordered probit model with a dummy intercept vector
Reinoso- Carvalho <i>et al.</i> (2020b)	1611	S	B	survey	Flavor liking, music–flavor match, chocolate (sweet, bitter, flavor strength, hardness, softness), music liking, buying intention, WTP	The multiresponse, multivariate, logit-normal regression model
Wang <i>et al.</i> (2020)	113	S	M	survey	Liking, bitter–sweet	Repeated-measures ANOVA
	137	S	M	survey	Liking, bitter–sweet, valence, arousal, sweet, bitter, sour, salty	Repeated-measures multivariate ANOVA, multiple mediation analysis using the Lavaan Structural Equation Modeling
Padulo <i>et al.</i> (2021)	75	S	W	food choice	Food most liked in pairs, reaction times, proportions of food selection	Repeated-measures ANOVA
Pantoja and Borges (2021)	55	L	B	survey	Taste/quality expectations, purchase intention	ANOVA
	101	L	B	survey	Taste/quality expectations, purchase intention	ANOVA
Rodríguez <i>et al.</i> (2021)	192	L	M	survey	Amused, calmed, comforted, curious, energetic, excited, friendly, good, joyful, pleased, responsible, disappointed, grumpy, restless, sad, rational, refreshing, sweet, bitter, alcohol	Repeated measures ANOVA
	400	L	B	survey		Multivariate ANOVA, Pearson correlations.
Speed <i>et al.</i> (2021)	456	S	W	survey	Aroma–color match, aroma–angularity shape match, aroma–pitch match, aroma–texture match, aroma liking, aroma naming	One-sample <i>t</i> -tests

Table A1.
(Continued)

Reference	N	L/S	Design	Measure- ment instrumen- tation	Measures	Data analysis techniques
Thaploo <i>et al.</i> (2021)	62 ^{i,vi}	S	W	survey	Aroma (intensity, pleasantness calmness), EEG segments of 5120 ms length, reaction time	Mixed-model repeated-measures ANOVA, multivariate ANOVA
Wang <i>et al.</i> (2021)	1819	L	W	survey	Sound–saltiness match	Chi-square test, one-way ANOVA, two-way ANOVA
Lin <i>et al.</i> (2022)	64	S	W	TDS, survey, CATA	TDS (sweetness, bitterness, cocoa, creamy, milky, roasted, and vanilla), emotion words, CATA (active, at ease, calm, energetic, enthusiastic, excited, interested, joy, pleasant, relaxed, satisfaction, annoyed, anxious, boredom, lonely, restless, tired, unable to concentrate, uneasy, unhappy), valence, arousal, dominance	One-way ANOVA, Correspondence Analysis, Canonical Variate Analysis, Partial Least Square Path Modelling (PLS-PM)
Motoki <i>et al.</i> (2022)	796	L	M	survey	Eating intention, valence, arousal, lyrics understanding, familiarity	ANOVA, SPSS PROCESS macro
Ramos-de-la-Peña <i>et al.</i> (2022)	51	S	W	survey	Sweet, salty, hot, bitter, creamy, oily, moisture, sticky, crunchy	ANOVA, PCA, Generalized
Vanden-berghe-Descamps <i>et al.</i> (2022)	107	S	W	survey	Harmony, familiarity, and hedonic of pairs, familiarity and hedonic of beer and soundtrack, separately.	Procrustes analysis chi-square, ANOVA, PCA
Guedes <i>et al.</i> (2023a)	106	S	W	survey	Sweet, like, likelihood of future consumption	ANOVA

ⁱ Elderly populationⁱⁱ Male populationⁱⁱⁱ Wine expert population^{iv} Female population^v Trained panelists^{vi} Young population

Table A2.

Description and characterization of auditory stimuli used in all experiments. Type of stimulus: So = song, St = soundtrack, To = tone, Am = atmospheric sound, Vo = voiceover, Pr = product-intrinsic sounds. Co/Ch: Co = composed, Ch = chosen. Intended gustatory correspondence: Bt = bitter, Sw = sweet, Sa = salty, So = sour, Sp = spicy, Sm = smooth, Ro = rough

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Crisinel and Spence (2012a)	Notes — in intervals of two tones from C2 to C6 — performed by piano, strings, woodwind, and brass, separately.	To	Co		To match a pitch and timbre with an aroma.	
Crisinel and Spence (2012b)	Notes — in intervals of two tones from C2 to C6 — performed by piano, strings, woodwind, and brass, separately.	To	Co		To match a pitch and timbre with an aroma	
Crisinel <i>et al.</i> (2012)	40 s of sweet and bitter soundtracks	St	Ch	Sw, Bt	A previous study demonstrated that these soundtracks are associated with sucrose and bitterness of caffeine, respectively	
North (2012)	Carmina Burana by Orff, Waltz of the Flowers by Tchaikovsky, Just Can't Get Enough by Nouvelle Vague, and Slow Breakdown by Michael Brook.	So	Ch		To correspond to emotional values: 'powerful and heavy', 'subtle and refined', 'zingy and refreshing', and 'mellow and soft'.	The four songs were classified between powerful/heavy, subtle/refined, zingy/refreshing, and mellow/soft ($N = 5$).

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Stafford <i>et al.</i> (2012)	News articles read by a male voice as shadow in a loop and Hardcore, 303 Freestyle, Jamie Ritmen — Scott Brown — Hardcore Heaven 4	Vo, So	Ch/Co		Hardcore was rated with the highest levels of distractibility and volume.	Drum&bass, house, hardcore, dubstep, and trance sounds were evaluated ($N = 10$).
Toffolo <i>et al.</i> (2012)	Flowing streams by M. Simmonds	So	Ch		Closest to Cassis aroma	15 females validated the song closest to Cassis aroma in terms of valence, potency, and activity. The songs were Flowing streams, Natural power, and Leaves in the wind, all by M. Simmonds, and Mecker by Ricardo Villalobos.
Xu and Kerr (2012)	Actual mastication sound	Pr	Co			
Stafford and Dodd (2013)	“Stress” by Justice at 85 bpm and 142 bpm	So	Ch		Contemporary music currently played in nightclubs.	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/ Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Stafford <i>et al.</i> (2013)	News articles read by a male voice as shadow in a loop and Hardcore, 303 Freestyle, Jamie Ritmen — Scott Brown — Hardcore Heaven 4. Both at 80 dB and 100 dB.	Vo, So	Ch		Hardcore was rated with the highest levels of distractibility and volume. High volume and low volume at those dB because they are the typical range experienced in a nightclub environment and roughly equivalent to the differences between the low and high levels in previous observational work. In a previous study, 10 participants evaluated drum and bass, house, hardcore, dubstep, and trance pieces of music.	
Fiegel <i>et al.</i> (2014)	5 mins of Air on the G string in 4 genres (classical, jazz, rock, hip-hop), each digitally arranged to be performed by a single and multiple performers.	So	Ch		The 4 genres were perceived as different (regardless single or multiple performers).	Musical pieces were rated similarity/dissimilarity ($N = 35$).
Seo <i>et al.</i> (2014)	5 s of Jingle bells, 5 s of Dallmayr prodromo coffee advertisement melody, 5 s of white noise.	So	Ch		To be congruent with cinnamon and coffee aromas, respectively.	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Velasco <i>et al.</i> (2014b)	1 s of high pitch (1100 Hz), 1 s of low pitch (70 Hz)	To	Ch		Previous research demonstrated that sour tastes were matched to high pitch and sweet tastes to both high and low pitches.	
Chambaron <i>et al.</i> (2015)	15 mins of radio program composed of different podcasts with a nutritional message embedded 3 times.	Vo	Ch			
Choi and Cheok (2015)	Bernard Bouchard's MIDI sound sample no. G10 and Bouchard's sound sample no. P03.	St	Ch		They were associated with happy and threatening emotions, respectively, in a previous study.	
Knoeflerle <i>et al.</i> (2015)	42 short chord progressions	St	Co		varying (in seven steps) in the sound's attack, discontinuity, pitch, roughness, sharpness, or speed	
	Four (sweet, sour, salty, bitter) 30 s soundtracks consisting of synthesizer bass, pads, piano, and arpeggiated synthesizer	St	Co	Sw, So, Bt, Sa	varying in pitch, sharpness, roughness, consonance, and discontinuity	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Kontukoski <i>et al.</i> (2015)	Sweet soundtracks (Trois Gymnopédies, No.2 Lent et triste by Erik Satie, Davidsbündeltänze, Op. 6, No. 18, Nicht schnell by Robert Schumann) and Sour soundtracks (Superscriptio by Brian Ferneyhough), fragments of transformed Argentinian tangos, musical sample by Mesz <i>et al.</i> (2012)).	So	Ch	Sw, So	The sounds were the most strongly associated with sweet and sour, respectively. Sweet: long duration, low dissonance, articulation, and loudness. Sour: high pitch, long duration, and high dissonance, brightness, and roughness.	16 sounds were associated with sweetness, sourness, saltiness, or bitterness ($N = 11$).
Reinoso-Carvalho <i>et al.</i> (2015a)	“Vem Morena, Vem” by Jorge Ben and ambient soundscape of the production kitchen	So, Am	Ch		Low sharpness, not either rapid or slow, balanced–low roughness. To resemble tropical conditions and elements of Latin American culture.	
Wang and Spence (2015a)	Debussy’s Jardin Sous la Pluie and Rachmaninoff’s Vocalise.	So	Ch		To match white wine and red wine, respectively.	
Wang and Spence (2015b)	Sweet and sour soundtracks	St	Ch	Sw, So	Both validated in a previous study	
Wang <i>et al.</i> (2015)	15 s of five bitter, five salty, seven sour, and seven sweet.	St	Ch	Bt, Sa, So, Sw	Published in a previous study composed as taste soundtracks	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Yan and Dando (2015)	Actual airplane cabin noise recordings delivered at 80–85 dB and low-volume ambient room noise.	Am	Co		Recreate cabin noise and control condition	
Endo <i>et al.</i> (2016)	Pseudo-chewing sound modified to amplify 250 Hz to 1 kHz frequency range	Pr	Co			
Kantono <i>et al.</i> (2016)	14 songs representative per musical genre	So	Ch		Musical genres from Rentfrow and Gosling (2003).	45 s from the beginning were enough per song ($N = 45$). Eight 2-h-day-music-listeners agreed one song representing each genre and provided musical keywords description
Reinoso-Carvalho <i>et al.</i> (2016)	60 s of “Oceans of Light” by The Editors.	So	Ch	So, Sw	To enhance the perceived sourness of the beer through high spectral balance, staccato articulation, syncopated rhythm, high pitch. To enhance the perceived sweetness through piano dominance.	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/ Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Wang and Spence (2016)	Two consonant (minor and major thirds) and two dissonant melodies (minor seconds) played by piano or both piano and trumpet	St	Co		To ensure that the experiment was focused on the high level concept of harmony, rather than on the specificities of a particular melody or instrument. Based on a study on infant perception of music consonance/dissonance reported.	Soundtracks were rated from very sour to very sweet and sour significantly matched to dissonant and sweet to consonant ($N = 30$)
Zhang <i>et al.</i> (2016)	One-hundred-twenty-five 10-s music clips from Chinese folk music database	So	Ch		Music experts selected 125 out of 200 clips to have a dominant emotion.	All the music was classified by valence/arousal evoked ($N = 20$).
Endo <i>et al.</i> (2017)	Pseudo-chewing sound during mastication	Pr	Co		To enhance the eating experience of elderly people obliged to follow texture-modified diets	
Guetta and Loui (2017)	8 violin clips with two versions of sweet, sour, salty, and bitter melody correspondence.	St	Co	Sw, So, Sa, Bt	To mimic the connotations of sweet, sour, salty, and bitter. Bitter triggers more low-pitched and legato notes; salty connotes staccato articulation; sour results in dissonance; and sweet is associated with consonant, slow and soft notes.	Melody–taste associations were validated ($N = 160$).

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Hathaway and Simons (2017)	Atmospheric sounds of a man making cookies in a domestic kitchen	Am	Co			
Reinoso-Carvalho <i>et al.</i> (2017)	One smooth and one rough soundtracks	St	Co	Sm, Ro	Soft sounds are correlated with long-consonant-legato notes, hard sounds are represented by short-dissonant-staccato notes, and the sound of the violin was rated as harder and drier than the sound of the flute. Two types of opening sounds.	Soundtracks were validated in creamy–rough and bitter–sweet ($N = 65$).
Spence and Wang (2017)	The sounds of screw-cap and cork opening.	Pr	Co			
Wang and Spence (2017)	One melody modified in 25 versions, ranging from C3 to C7 (5 pitches), from 60 to 380 bpm (5 tempos), and with piano or strings.	St	Co			30 people validated all sounds. High tempo and pitch were associated with cold beverages

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Wang <i>et al.</i> (2017a)	Spicy soundtrack, sweet soundtrack, white noise.	St	Co	Sp, Sw	Spicy (high pitch, fast tempo, and a distorted timbre): Composed based on the sound attributes found to be spicy in Expl. Sweet (high pitch, legato articulation, and consonant harmony): Composed based on literature.	Both soundtracks were validated in sweet, spicy, sour, and salty ($N = 110$).
Wang <i>et al.</i> (2017b)	Spicy soundtrack, sweet soundtrack, white noise.	St	Co	Sp, Sw	Both sweet and sour soundtracks were validated in previous studies.	Image plus sound were validated online ($N = 110$).
	Spicy soundtrack	St	Co	Sp		
	30 s of concatenating sweet followed by sour soundtracks and concatenating sour followed by sweet soundtracks.	St	Co	Sw, So		
Kantono <i>et al.</i> (2018)	Most liked, least liked, and neutral song for each participant.	So	Ch		A set of songs broadly representing most music genres according to Kantono <i>et al.</i> (2016)	Each participant rated the most liked, least liked, and neutral music.
Rahne <i>et al.</i> (2018)	White noise at 70 dB	Am	Co			

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Roque <i>et al.</i> (2018)	Four 2-s sounds of liquid being poured into glass (liquid, liquid + ice cubes, carbonated, carbonated + ice cubes) with pitch normalized	Pr	Co			
Sinesio <i>et al.</i> (2018)	Countryside, leaves' rustling, birds chirping atmospheric sounds	Am	Ch		To reproduce a rural scenario and produce closeness to the production site perception	
Wang and Spence (2018b)	Sweet and sour soundtracks	St	Ch	Sw, So	Both sweet and sour soundtracks were validated in two published studies.	
	Two abstract (aleatoric) soundscapes	St	Co		One is sparsely textured and staccato, and the other is less sparse with overlapping legato woodwind lines. They were associated with different tastes (sour-sweet) and different pleasantness in a previous study.	
Wang and Spence (2018c)	Consonant and dissonant melodies	St	Ch			
Ziv (2018)	Prelude 1 in C major, from Bach's Well-Tempered Clavier, and Walking by Guillermo Balvin.	So	Ch		Most pleasant (tonal music) and least pleasant (atonal music), respectively. Avoid varying timbre, tempo, instrumentation.	Six classical tonal and atonal songs were validated for piano on pleasantness and likability ($N = 20$)

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Ba and Kang (2019)	Prelude 1 in C major, from Bach's Well-Tempered Clavier, Walking by Guillermo Balvin, a piano version of the prelude of Bach's suite for cello in G, and Schoenberg's suite for keyboard (suite for claviers) op. 25.	So			Most similar pleasantness rating to the respective tonal and atonal music in Experiment 1.	5 pieces were validated in pleasantness ($N = 16$).
	40 s of birds, conversation, and traffic at low, mid, high volume.	Am	Co		Natural, anthropogenic, and mechanical sounds, respectively, common in urban environments.	Subjects previously evaluated subjective loudness.
	100 Hz tone and 1000 Hz tone	To	Ch		Low vs high pitch	
Burzynska et al. (2019)	Wolfgang Amadeus Mozart's Flute and Harp Concerto in C Major, K.299/297c, 2nd movement and Johann Sebastian Bach's Brandenburg Concerto No. 3 in G Major, BWV 1048, 1st movement.	So	Ch		Mozart's song is down tempo (low arousal), Bach's song is up tempo (high arousal).	Sounds were assessed arousal level by six items ($N = 34$). The pieces had different arousal qualities and controlled by pleasantness.

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/ Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Fiegl <i>et al.</i> (2019)	3 mins of “Air on the G string” by Bach modulated to high (880–2093 Hz) <i>vs</i> low (110–261 Hz) pitch, slow (40 bpm) <i>vs</i> fast (92 bpm) tempo, and quiet (47–67 dB) <i>vs</i> loud (61–90 dB) volume, separately.	So	Ch			
Hauck and Hecht (2019)	11 mins of “Waltz of the Flowers” by Tchaikovsky and “Three pieces for orchestra (March)” by Berg	So	Ch		“Waltz of the Flowers” had the highest global liking score. “Three pieces for orchestra (March)” differed significantly from “Waltz of the Flowers” in 14 out of 16 dimensions.	5 classical music pieces were rated on global liking, sweet, sour, salty, bitter, foul, floral, aromatic, fruity, lively, gloomy, harmonic, light, zingy/refreshing, powerful/heavy, subtle/refined, mellow/soft, likewise ($N = 18$).
Höchenberger and Ohla (2019)	Bitter-associated soundscape, sweet-associated soundscape, 5 s of sound snippets bitter/sweet associated.	St	Ch	Bt, Sw	Bitter (low-pitch trombone tunes) and sweet (high-pitch piano tunes) associations based on Crisinel <i>et al.</i> (2012).	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Hsu and Chen (2019)	Two French songs (La Vie en Rose and J'attendrai), two Spanish songs (La Paloma and Bésame Mucho), and classical music.	So	Ch			
Jutras <i>et al.</i> (2019)	Actual mastication sound.	Pr	Co			
Kantono <i>et al.</i> (2019)	Fourteen 45-s songs (one per genre)	So	Ch		The songs were classified into liked, disliked, or neutral based on their liking validated in a previous study.	
Lin <i>et al.</i> (2019)	Café, fast food restaurant, bar, food court, and park atmospheric sounds.	Am	Co		Representative locations where food is frequently consumed	
Liu <i>et al.</i> (2019)	Coffee and construction atmospheric sounds.	Am	Co		Visual, gustatory, and olfactory in/congruence	
Luca <i>et al.</i> (2019)	"Cannot stop the feeling" by Justin Timberlake and "Turkish March" by Wolfgang Amadeus Mozart.	So	Ch		One trendiest song (peppy lively music) and one most popular classical song (fine music).	
Mathiesen <i>et al.</i> (2019)	Sweet and bitter soundtracks	St	Co	Sw, Bt	Sweet (high-pitch, slow clarinet/piano, string pad, consonant, legato) and bitter (low-pitch, brass drone, grainy/rhythmical cello) associations were based on previous studies.	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/ Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Mitchell <i>et al.</i> (2019)	7 s of effervescent beverage sound	Pr	Co			Sound was bench-tested to ensure it was representative of effervescence.
Motoki <i>et al.</i> (2019)	Three 8-s advertisements for different foods in original pitch, high pitch (20% and 3.16 semitones), and low pitch (−30% and −6.17 semitones).	Vo	Co		To check whether soundtrack crossmodal correspondences are the same for voiceovers.	
Peters Rit <i>et al.</i> (2019)	Eight 3-s high-tempo music clips and Eight 3-s low-tempo music clips.	So	Ch		To represent a broad range of emotional arousal based on experimenter intuition.	24 aromas and sounds were rated to select 8 lowest and 8 highest on arousal ($N = 10$).
Reinoso-Carvalho <i>et al.</i> (2019)	Nocturne by Chopin and Mors Praematura by Curry	So	Ch		Positive and negative emotion, respectively.	Nocturne, Porro Sabanero, Adagio for Strings, and Mors Praematura were emotionally validated through PANAS scale ($N = 263$).
	Nocturne by Chopin, Mors Praematura by Curry, Porro Sabanero by Bermudez.	So	Ch		Porro Sabanero: folkloric and very incongruent with beer.	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Wang <i>et al.</i> (2019a)	Oak-ageing soundtrack	St	Co	Bt	To reflect the flavor qualities of bitter/woody. Cello, woodblocks, temple bells, some low frequency percussion, a sustained woodwind texture, and a low drone sound comprised of several elements (including voice and string instruments).	
Wang <i>et al.</i> (2019b)	45 s of Brian Eno's "Discreet Music" and Mussorgsky's "A Night on Bald Mountain"	So	Ch		Discreet Music: high fruitiness, low tannin, static, 70 bpm, consonant harmony. A Night on Bald Mountain: dynamic changes in orchestration/register/loudness, 121 bpm, consonant and dissonant harmonies.	Two songs were different in fruitiness and tannin levels ($N = 19$).
Wang <i>et al.</i> (2019c)	60-s sweet-associated and bitter-associated sounds	St	Ch	Sw, Bt		
Xu <i>et al.</i> (2019)	Café atmospheric sound, café with bird sounds, café with forest sounds, café with machine sounds.	Am	Ch		Sweet and bitter correspondence tested in a previous study.	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Bravo-Moncayo <i>et al.</i> (2020)	1 min of baseline urban atmospheric noise (71 dBA), passive-controlled noise (34 dBA, lower mid-high frequencies), and active-controlled noise (30 dBA).	Am	Co			
Covaci <i>et al.</i> (2020)	Original video sound, sound adjusted to 328 Hz (high-pitch), and sound adjusted to 41 Hz (low-pitch).	Am	Ch		High and low pitch conditions.	
Fibrianto <i>et al.</i> (2020)	Basic tone F for symphony no. 6 k.43 at 3 different duration times.	So	Ch			
Mesfin <i>et al.</i> (2020)	Video sound at 328 Hz (high pitch) and 41 Hz (low pitch).	Am	Co		High and low pitch conditions.	
Nivedhan <i>et al.</i> (2020)	1 positive valence and 1 negative valence music excerpts.	So	Ch		1 positive valence and 1 negative valence music excerpts. Both negative emotional arousal.	76 people determined the elicited emotions of colors and 8 music cues.

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Peng-Li <i>et al.</i> (2020)	Sweet and salty soundtracks balanced between randomness and predictability	St	Co	Sw, Sa	Sweet: C-major arpeggio, high pitch, legato, consonant, slow tempo, warm timbre, and clarinet melody. Salty: minor chords, staccato and fast tempo. Features validated in a previous study.	
Qi <i>et al.</i> (2020)	Twenty-eight 1.5-s notes at different pitches from erhu, guzheng, dizi, or yunluo instruments.	To	Co		Nonwestern sounds	
Reinoso-Carvalho <i>et al.</i> (2020a)	Positive-emotion, negative-emotion, sweet, and bitter songs.	St, So	Ch	Sw, Bt	To evaluate if emotional music trigger more effective seasoning than crossmodally corresponding music. Validated in a previous study.	
Reinoso-Carvalho <i>et al.</i> (2020b)	Four 1-min songs: soft, rough, positive emotion, negative emotion.	St, So	Ch	Sm, Ro	To avoid priming by any style, positive and negative songs are classical. Validated in a previous study.	4 songs were online tested in sweet, bitter, hard, soft, active, attentive, enthusiastic, excited, inspired, distressed, irritable, nervous, scared, upset ($N = 135$)
Wang <i>et al.</i> (2020)	30-s sweet and bitter soundtracks	St	Ch	Sw, Bt	Both validated in a previous study	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/ Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Padulo <i>et al.</i> (2021)	Twelve 15-s soundtracks (4 salty, 4 sweet, 4 neutral)	St, Am	Ch	Sa, Sw	Salty (medium pitch, staccato, dissonant) and sweet (high pitch, legato, consonant) soundtracks were validated in previous studies. Sensory neutral: bird chirping, driving car, environmental.	
Pantoja and Borges (2021)	Smooth jazz backing track and “One More Night” by Maroon 5, both at 72 bpm and 110 bpm.	So	Ch		Manipulated the tempo of two songs following research on tempo (slow: 72 BPM; fast: 110 BPM)	Sounds were validated as significantly fast and slow, respectively, regarding the tempo.
Rodríguez <i>et al.</i> (2021)	4 sound pressure level and pitch manipulations of pouring and opening sounds of a beer can and bottle (16 sounds). Bottle pouring sound high pitch/ high sound pressure, low pitch, and low sound pressure.	Pr	Ch		Sounds were validated in premiumness in a previous study.	
Speed <i>et al.</i> (2021)	2 s of 330 Hz, 262 Hz tones.	To	Ch		High and low pitch conditions.	
Thaploo <i>et al.</i> (2021)	Low pitch (~500 Hz) and high pitch (~700 Hz).	To	Ch		High and low pitch conditions.	

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/ Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Wang <i>et al.</i> (2021)	Thirty-six 2 to 10 s of sound clips representing articulation, tempo, length of attack, length of decay, consonance, harmonic complexity, randomness, syncopation, roughness, reverberation, mode, valence, and arousal.	St	Co		The sounds were composed to ensure that the only variation between each clip in the set was in one specific musical attribute.	
Lin <i>et al.</i> (2022)	45 s of subjective liked, disliked, pleasant, and unpleasant music.	So	Ch		45 s were the optimum time for the rating of the pleasantness of ice cream. Music was selected and categorized for each genre according to the Apple iTunes music classification system. Participants rated fourteen different music genres to find the liked, disliked, pleasant, and unpleasant music.	The song that best represented each genre was validated ($N = 60$). Fourteen sounds best represented pleasant and unpleasantness ($N = 68$).

Table A2.
(Continued)

Reference	Stimulus description	Type of stimulus	Co/Ch	Intended gustatory correspondence	Rationale	Stimuli pre-validation
Motoki <i>et al.</i> (2022)	Five 30-s representative songs of Jazz, Classical, Rock/Metal, and Hip-hop.	So	Ch		They were selected in a previous study on music genre and food preferences. Rock/Metal to make clear distinction to others. Sounds were selected with the assistance of the classification of music genres in Apple music.	
Ramos-de-la-Peña <i>et al.</i> (2022)	Sounds of 7 videos to evoke heat, sadness, coldness, height/fear, pleasure, speed/fear, relaxation.	Am	Co		To evoke corresponding emotions.	
Vandenbergh-Descamps <i>et al.</i> (2022)	20 s of songs from Ireland, USA, Senegal, and India.	So	Ch		The most typical soundtrack from where the beers originate	Typicality of 3 movie soundtracks/songs from each corresponding country were evaluated ($N = 20$ French (nonnative listeners)).
Guedes <i>et al.</i> (2023a)	30 s of “Fruit of Lore” (mid–high pitch, guitar/woodwind, legato, 80 bpm) and “Walk of Destiny” (152 bpm, staccato, strings/percussion, mid pitch)	So	Ch	Sweet	High sweetness (was evaluated in a previous study as sweet by 80.52% of the sample in the norming study) and low sweetness (was rated as sweet by only 7.02% of participants), respectively, with approximate valence ratings (high) and unfamiliar	

Table A3.

Description and characterization of food stimuli used in all experiments. Type of stimulus: Fo = food, Be = beverage, Ac/Ex: Ac = actual experience, Ex = expectation. Pr/Ch: Pr = produced, Ch = chosen. Basic taste association: Sw = sweet, Bt = bitter, So = sour, Sa = salty, Um = umami

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Crisinel and Spence (2012b)	Fo	5 g of dark chocolate, 5 g milk chocolate, 6.7 g of marzipan-filled chocolate.	Ac	Ch		Commercially available, proven crossmodal associations, variety of pleasantness (dark chocolate), almond as strong pleasant and unpleasant.	
Crisinel <i>et al.</i> (2012)	Fo	0.3 g of cinder toffee	Ac	Pr			
North (2012)	Be	Red wine and white wine.	Ac	Ch			
Stafford <i>et al.</i> (2012)	Be	25 ml of cranberry/vodka mixes at 0%, 1.9%, 3.9%, 5.6%, 7.5% of alcohol, respectively.	Ac	Pr	Sw, Bt	Cranberry for not easily identification of differences in taste (sweet/bitter) and strength.	Tesco smooth cranberry, Schweppes Indian tonic water, and Tesco pure orange juice, were rated ($N = 10$)
Xu and Kerr (2012)	Fo	Vacuum belt dried chips and deep-fat fried chips.	Ac	Pr		A low-fat version (vacuum belt dried) of an unhealthy product.	
Stafford and Dodd (2013)	Be	Two sips of vodka.	Ac	Ch	Bt, Sw	variability in terms of bitter/sweet.	5 females validated taste and pleasantness of the 5 most consumed drinks.
Stafford <i>et al.</i> (2013)	Be	25 ml of cranberry/vodka mixes at 0%, 1.9%, 3.9%, 5.6%, 7.5% of alcohol, respectively.	Ac	Pr		Most appropriate levels of alcohol and mixer.	

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Fiegel <i>et al.</i> (2014)	Fo	Milk chocolate and red bell pepper.	Ac	Ch		Chocolate: emotional. Bell pepper: nonemotional.	Foods consumed when mood is altered were chosen from 29 foods ($N = 150$). Most chosen: chocolate, least chosen: bell pepper.
Velasco <i>et al.</i> (2014b)		Packaging image of sour/ sweet hypothetical products.	Ex		So, Sw		
Chambaron <i>et al.</i> (2015)	Fo	'Pain of chocolat' aroma, as well as likely starter (cold meat or grated carrots), likely main course (lasagna or salmon with green risotto), and likely dessert (waffle or applesauce). Bitter, sweet, sour, and salty terms	Ex	Pr		The aroma was selected for being sweet-fatty. The food was selected to recreate a common meal choice situation, and to compare high-energy density vs low-energy density.	The aroma was validated for not being detectable ($N = 10$). Expected liking was similar among dishes ($N = 21$).
Knoefler <i>et al.</i> (2015)			Ex		Sw, So, Bt, Sa		
Kontukoski <i>et al.</i> (2015)	Be	Mango, orange, grapefruit, lemon, and pineapple juices, and liquid honey.	Ex	Pr	Sw, So	To select the same amount of sweet and sour ingredients with similar color (yellow) and structure (clear).	28 foods were classified into sweet, sour, bitter, and salty ($N = 8$).
Reinoso-Carvalho <i>et al.</i> (2015a)	Fo	A chocolate sample named 'Brasil'.	Ac	Pr		Colored mass based on white chocolate, together with a mixed filling of lime jelly and coriander 'ganache'.	

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Wang and Spence (2015a)	Be	White wine and red wine.	Ac	Ch		To be very different to facilitate musical matching.	
Wang and Spence (2015b)	Be	10 ml of unflavored vodka, raspberry vodka, and citron vodka.	Ac	Ch		To correspond to atmospheric conditions.	
Wang <i>et al.</i> (2015)		Sweet, sour, bitter, and salty terms	Ex		Sw, So, Bt, Sa		
Yan and Dando (2015)	Be	2 oz of sucrose, NaCl, citric acid, MSG, and quinine HCl samples of aqueous food grade solutions from Sigma Aldrich delivered in distilled deionized water at three level of concentrations	Ac	Pr	Sw, So, Sa, Bt, Um	For covering five basic tastes.	
Endo <i>et al.</i> (2016)	Fo	Five nurse care meals.	Ac	Pr		According to Japan Care Food Conference guidelines, classified as category 2 or 3. Meals had been cooked until very soft.	
Kantono <i>et al.</i> (2016)	Fo	Dark, bittersweet, and milk chocolate gelati.	Ac	Pr	Bt, Sw	Having bitter, transitional, and sweet samples follows the influence of musical notes on pleasantness ratings of three gelati.	

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Reinoso-Carvalho <i>et al.</i> (2016)	Be	“Salvation” beer.	Ac	Pr		5.3% of alcohol. Inspired by “In dream” album and to be disruptive.	
Wang and Spence (2016)	Be	Orange/grape/apple, grape/ apple/pear, and grape/peach juice mixtures.	Ac	Pr		To minimize the possibility that the participants would guess the nature of the juice.	
Endo <i>et al.</i> (2017)	Fo	Three minced-like and three puréed-like nurse care meals.	Ac	Ch		To measure the influence of texture homogeneity.	
Guetta and Loui (2017)	Fo	4 custom-made chocolates: sweet (61% semisweet cacao chocolate), sour (1.5 tablespoons of white distilled vinegar per 10 ounces of 61% semisweet cacao), salty (1 teaspoon of salt per 8 ounces of 61% semisweet cacao chocolate), and bitter (100% bittersweet cacao).	Ac	Pr	Sw, So, Sa, Bt	To adapt them to the four basic tastes (sweet, sour, salty, and bitter).	Flavor match and strength similarity was validated ($N = 19$).
Hathaway and Simons (2017)	Fo	4 brands of chocolate chip soft-style cookies.	Ac	Ch		They differed enough in quality to obtain discriminating hedonic responses, were similar enough in appearance to prevent memorizing samples, and were all store-bought.	Participants validated differences in quality to obtain discriminating hedonic responses.

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Reinoso-Carvalho <i>et al.</i> (2017)	Fo	4 chocolates with different cocoa content.	Ac	Pr		To keep the color of the chocolates as similar as possible.	Professional chocolatiers determined which combination of cacao would be appropriate to use.
Spence and Wang (2017)	Be	Dry red wine and red wine.	Ac	Ch		The two wines were chosen to be similar but not identical and hence ensure a meaningful difference in each of the two parts of the experiment.	
Wang and Spence (2017)	Be	Distilled water at 5° C, 21° C, and 45° C.	Ac	Ch			
Wang <i>et al.</i> (2017a)	Fo	Spicy-sweet dish.	Ac	Pr	Sw	Contained both sweet and spicy components. To be spicy.	Capsaicin was added until taste was around 5/10 ($N = 3$).
	Fo	Medium spicy salsa mixture.	Ac	Pr		Different heat levels. Enhancing participants' expectations. The same brand was used to maximize consistency in taste between the two different levels of spiciness.	
	Fo	Mild Salsa and Hot Salsa.	Ac	Ch			

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Wang <i>et al.</i> (2017b)	Be	15 ml of off-dry wine.	Ac	Pr		Low alcohol (7.5%), crisp acidity, a medium level of residual sugar (73.3 g/l), aromas of white peach, apricot, and citrus, and citrus flavors with a good balance between acidity and sweetness.	
Kantono <i>et al.</i> (2018)	Fo	Bittersweet chocolate gelato.	Ac	Ch	Bt–Sw		
Rahne <i>et al.</i> (2018)	Od	16 varying dilutions of n-butanol as odorants as well as sucrose, NaCl, citric acid, quinine HCl filter paper strips as tastants	Ac	Pr	Bt, Sw, Sa, So	The odorants to evaluate olfactory threshold from one of each triplet containing the solution. The tastants, to fit the four basic tastes: sweet, sour, salty, and bitter	
Roque <i>et al.</i> (2018)	Be	Image of carbonated beverage.	Ex				
Sinesio <i>et al.</i> (2018)	Fo	Tomato salad and wild rocket salad (both 24 h postharvest and 7 days storage)	Ac	Pr		They differed in freshness without inducing macroscopic decay in quality.	Perceivable difference in quality but no evident deterioration was validated.
Wang and Spence (2018b)	Be	Two white blends wines 2014 and 2015.	Ac	Ch		The wines were chosen to be different but similar, so that the participants would not necessarily assume the wines to be the same.	

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Wang and Spence (2018c)	Be	White wine 2012 and 2013.	Ac	Ch			
	Be	15 ml of a juice mixture between white grapefruit juice, orange juice, and apple juice.	Ac	Pr		To minimize the possibility that the participants would guess the nature of the juice and hence perhaps feel overconfident.	
Ziv (2018)	Fo	3 cm diameter vanilla cookies	Ac	Pr			69 wine evaluation trainees rated
Burzynska et al. (2019)	Be	20 ml of two red wines.	Ac	Ch		A similar medium body, but different levels of acidity and aromas. One wine has 13.5 abv, 3.82 ph, 4.9 g/l acidity, 1.2 g/l residual sugar; and the other has 14.1 abv, 3.50 ph, 5.7 g/l acidity, 2.7 g/l residual sugar.	wines across 58 wine attributes, covering color, aroma, flavor, mouthfeel, aftertaste, and liking.
Cho et al. (2019)	Fo	5 g of chocolate chip cookies.	Ac	Ch			
Fiegel et al. (2019)	Fo	Milk chocolate and red bell peppers.	Ac	Ch		In a previous study, chocolate and bell peppers were considered to be emotional and nonemotional foods, respectively.	
Hauck and Hecht (2019)	Be	50 ml of red wine, white wine, sugar water solution, and citric-acid water solution.	Ac	Ch/ Pr		The solutions were strong enough to produce a recognizable taste, but faint enough to not risk the provocation of disgust or nausea.	

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Höchenberger and Ohla (2019)	Fo	Home-made bittersweet cinder toffee.	Ac	Pr	Bt–Sw	Taste equally bitter and sweet.	
Hsu and Chen (2019)	Be	30 ml of French 2015 white wine and Spanish 2015 red wine.	Ac	Ch		Rated 3.3 on vivino.com and costed \$15.	
Juras <i>et al.</i> (2019)	Fo	Crispy foods (potato chips, snack salty pretzel, cereal bar, carrot, and apple) and soft food (soft pretzel, soft granola bar coated with chocolate, soft cookie, banana, and tomato).	Ac	Ch			
Kantono <i>et al.</i> (2019)	Fo	Bittersweet chocolate gelato	Ac	Pr	Bt–Sw		
Lin <i>et al.</i> (2019)	Fo	Chocolate gelato	Ac	Pr		Chocolate has been reported to be an emotional food that is influenced by background music. To melt slowly in the mouth with no strong bitter after taste sensations persisting before the next sample was tasted.	

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Liu <i>et al.</i> (2019)	Be	3 cold brew coffees made from different beans.	Ac	Pr		One bean is intense and smoky in character, another is smooth and deep; and the other has notes of cocoa, toasted almond and caramel.	
Luca <i>et al.</i> (2019)	Be	White wine and red wine	Ac	Ch			
Mathiesen <i>et al.</i> (2019)	Be	Coffee beverage	Ac	Pr			
Mitchell <i>et al.</i> (2019)	Be	Molson Exel beer	Ac	Ch		A dealcoholized (<0.5%) beer was chosen so that multiple samples could be swallowed rather than expectorated, thus avoiding potential impairment in judgement from alcohol accumulation and recruiting taste receptors in the pharynx.	
Motoki <i>et al.</i> (2019)	Be, Fo	Cream puff, lemon juice, and black coffee audio advertisements.	Ex		Sw, So, Bt	To correspond to basic tastes sweet, sour, bitter.	
Reinoso-Carvalho <i>et al.</i> (2019)	Be	100 ml of Zimnebir beer	Ac	Ch		Belgian bitter-dry pale lager, with 6% alcohol (malty, with fine-salient bitterness).Most popular Belgian pale lager beer.	

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
	Be	Jupiler beer	Ac	Ch		Belgian pale lager beer (5.2% alcohol)	
	Be	Chimay Blue beer	Ac	Ch		Very different from the beers used in the previous experiments. Belgian strong dark ale, with 9% alcohol, dense, strong caramel flavored, and smooth palate.	
Wang <i>et al.</i> (2019a)	Be	2 white 2015 and 2014 wines and 2 red 2014 wines.	Ac	Ch			Hedonic attitude towards the wine was evaluated ($N = 14$).
Wang <i>et al.</i> (2019b)	Be	Red wine	Ac	Ch			
Wang <i>et al.</i> (2019c)	Be	Apple elderflower fruit beverage with no or two different pomegranate aroma concentrations	Ac	Pr	So	Beverage was relatively sour and somewhat unfamiliar. Aroma was unfamiliar.	
Xu <i>et al.</i> (2019)	Fo	50 g of chocolate ice-cream.	Ac	Pr			
Bravo-Moncayo <i>et al.</i> (2020)	Be	Heat brew coffee from blended Arabica green beans	Ac	Pr		Medium roasted, and at medium grinding, harvested from the Ecuadorian highlands.	
Fibrianto <i>et al.</i> (2020)	Be	Green tea with 1-, 3-, and 5-min brewing times, respectively, at 80° C.	Ac	Ch			

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Nivedhan <i>et al.</i> (2020)	Be	4 black cold brew coffees with varying sucrose levels (0%, 2.5%, 5%, 7.5%).	Ac	Pr			
Peng-Li <i>et al.</i> (2020)	Fo	8 sweet and 8 salty food images varying in fat content	Ex		Sw, Sa		
Qi <i>et al.</i> (2020)		Sour, sweet, bitter, salty, and umami terms.	Ex		So, Sw, Bt, Sa, Um		
Reinoso-Carvalho <i>et al.</i> (2020a)	Fo	1 g of milk and dark chocolates	Ac	Ch			
Reinoso-Carvalho <i>et al.</i> (2020b)	Fo	Milk and dark chocolates	Ac	Ch			
Wang <i>et al.</i> (2020)	Fo	3 g of 70% Lindt bittersweet chocolate	Ac	Ch	Bt, Sw	Fairly complex taste, sweet–bitter balance, and commercial availability.	
Padulo <i>et al.</i> (2021)	Be, Fo	128 food images included (high-calorie sweet, low-calorie sweet, high-calorie salty, and low-calorie salty).	Ex		Sw, Sa		

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Pantoja and Borges (2021)	Fo	Burgers and pizza visual cues.	Ex				
Rodríguez <i>et al.</i> (2021)	Be	Refreshing, sweet, bitter, alcohol terms.	Ex		Sw, Bt		
	Be	Nonalcoholic-Beer bottle pouring and Nonalcoholic-Beer can pouring images Saltiness expectation	Ex				
Wang <i>et al.</i> (2021)			Ex		Sa		
Lin <i>et al.</i> (2022)	Fo	Bittersweet chocolate ice cream	Ac	Pr	Bt, Sw	Formulated using a mixture of cream (51.54%), milk (15.38%), sugar (15.38%), and cocoa powder (7.69%).	Differences between food names were tested ($N = 40$).
Motoki <i>et al.</i> (2022)	Be, Fo	4 healthy (low-fat, low-calorie) savory, 4 healthy sweet, 4 indulgent savory, and 4 indulgent sweet foods.	Ex		Sw		
Ramos-de-la-Peña <i>et al.</i> (2022)	Fo	Tomato ketchup and ranch dressing.	Ac	Ch		To avoid interference with other factors, and to focus only on their comparison.	

Table A3.
(Continued)

Reference	Type of stimulus	Stimulus description	Ac/ Ex	Pr/ Ch	Basic taste association	Rationale	Stimulus pre-validation
Vanden-berghe-Descamps et al. (2022)	Be	20 ml of O'Hara (Irish beer), Bud (American beer), Gazelle (Senegalese beer), and Kingfisher (Indian beer).	Ac	Ch		To identify beers issued from four different parts of the world, typical from their country and different from each other. 20 ml to assure the blood-alcohol maximum level.	Typicality of 18 beers from 12 different countries were tested ($N = 20$).
Guedes et al. (2023a)	Fo	3 cm diameter piece of baby carrot and baby cucumber. 2.5 cm diameter piece of two versions of commercially available chocolate chip mini cookies with different sugar content	Ac	Ch	Sw	Baby carrot and baby cucumber: According to the Table of Food Composition 5.0, raw baby carrots present an average of 4.8 g of sugar per 100 g, whereas raw cucumbers present an average of 1.6 g of sugar per 100 g. Cookies: According to the manufacturer, the standard cookies contain about 30% sugar, whereas the 0% sugar mini cookies present <0.5 g sugar per 100 g of product.	

Table A4.

Description and characterization of aroma stimuli used in all experiments

Reference	Chemosensory stimulus description	Rationale	Stimulus pre-validation
Crisinel and Spence (2012a)	Almond, apple, apricot, blackberry, caramel, cedar, dark chocolate, cut hay, green pepper, honey, lemon, liquorice, mushroom, musk, pepper, pineapple, raspberry, smoked, vanilla, and violet aromas.	Aromas commonly found in wine. To cover a wide olfactory range.	
Toffolo <i>et al.</i> (2012)	Cassis.	A previous study indicated it to be a neutral aroma in valence, potency, and activity.	
Seo <i>et al.</i> (2014)	10% diluted cinnamon aroma, 10% diluted coffee aroma.		
Choi and Cheok (2015)	Bodyshop's Love (orient note) and Kousaido's Citron (citrus note).	Love smells sweet and warm and Citron is stronger and sometimes gives an uncomfortable feeling.	
Zhang <i>et al.</i> (2016)	Tea, garlic, fennel, burnt smell, coffee bean, smoke, mildew, cream, almond, lemon, grass, menthacamphorate, rodent repellent, sandalwood, watermelon lollipop, lavender, soil, whiskey, roasted sweet potato, and mothball.	4 food major students found 30 aromas to evoke different emotions throughout 4 quadrants. 20 were selected to keep the accuracy of the sense.	Aromas were classified by valence and arousal of emotions evoked ($N = 15$).
Ba and Kang (2019)	Lilac, osmanthus, coffee, and bread aromas, at low, mid, and high levels of concentration.	Plan aromas because they are from common varieties of urban greening; and food aromas because they are single, unmixed as opposed to varying.	Familiarity rating and subjective intensity of aroma stimuli were measured.
Peters Rit <i>et al.</i> (2019)	30 ml cucumber, leather, green tea, grapefruit, milk, coconut, soap, jasmine, garlic, gasoline, mint, vanilla, coffee, orange, valerian root, and chamomile essential oils.	Essential oils for representing a broad range of arousal (first 8 have high-arousal and last 8 have low-arousal), based on experimenter intuition and a panel pre-validation.	24 aromas and sounds were rated on arousal ($N = 10$). 8 aromas rated lowest and 8 rated highest were selected.

Table A4.
(Continued)

Reference	Chemosensory stimulus description	Rationale	Stimulus pre-validation
Covaci <i>et al.</i> (2020)	Rosemary, bergamot, lilial, clear lavender (low intensity), lavender (high intensity), lemon, and raspberry.		
Mesfin <i>et al.</i> (2020)	Bergamot, lilial, clear lavender (low intensity), lavender (high intensity), lemon, and raspberry.	To crossmodally correspond with visual content.	
Speed <i>et al.</i> (2021)	Sniffing sticks with scents of caramel, lemon, onion, menthol, and raspberry.	Based on reported associations between these aromas and other sensory modalities by previous studies.	
Thaploo <i>et al.</i> (2021)	Peppermint oil, vanilla, musk, and orange.	Aromas typically rated as pleasant but that cover different areas of the olfactory space.	10 experienced individual rated aroma intensities to adjust them to be approximately equal.

Table A5.

Auditory–chemosensory associations disaggregated in basic attributes of perception and personal preference validated throughout all experiments. The associations that were not disaggregated are reported in the ‘Other auditory–chemosensory associations’ column

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Crisinel and Spence (2012a)	Piano		Pleasant aroma	
	Brass		Unpleasant aroma	
	High pitch	Fruity		
	Brass	Acrid		
	Piano	Low acrid		
	Piano	Fruity		
	Woodwind	Low spicy		
	Piano	Low spicy		
Crisinel and Spence (2012b)	Piano	Low bitter		
	Piano	Sweet		
	Low pitch		Dark chocolate preference	
	Mid pitch		Marzipan chocolate preference	
	High pitch		Milk chocolate preference	
	High pitch	Sharp		
	High pitch	Light		
	High pitch	Crunch		
	High pitch	Clean		
	High pitch	Fresh		
	Piano		Pleasant	
	Strings		Unpleasant	
	Woodwind		Unpleasant	
Crisinel <i>et al.</i> (2012)	Trombone	Bitter		
	Low pitch	Bitter		
	Filtered noise	Bitter		
	Piano	Sweet		
	High pitch	Sweet		
North (2012)				Powerful/subtle/zingy/mellow music — powerful/subtle/zingy/mellow wine
Stafford <i>et al.</i> (2012)	High distractibility	Sweet		Shadow with music with high distractibility — low
	High volume	Sweet		discrimination of alcohol
	High distractibility	Low bitter		strength / higher alcohol
	High volume	Low bitter		strength in 0% alcohol drink
	High distractibility	Alcohol		
Toffolo <i>et al.</i> (2012)	Flowing streams song		Cassis	

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Xu and Kerr (2012)	Low peaks	Crunch		
	Low sound energy	Crunch		
	Low peaks		Chips preference	
	Low sound energy		Chips preference	
Stafford and Dodd (2013)				Music — less consumption time
Stafford <i>et al.</i> (2013)	High distractibility	Alcohol		Shadow with music with high distractibility — alcohol strength / low discrimination of alcohol strength.
	Low volume	Low sweet		
Fiegel <i>et al.</i> (2014)	Jazz		Vegetable preference	Music familiarity/stimulation/pleasantness — vegetable/chocolate intensity
	Jazz		Chocolate preference	
Seo <i>et al.</i> (2014)	Pleasant	Intensity		
	Christmas carol		Cinnamon matching	
	Piano		Coffee matching	
	Legato		Coffee matching	
Velasco <i>et al.</i> (2014b)	Triple meter		Coffee matching	
	High pitch	Sour		
Chambaron <i>et al.</i> (2015)	Low pitch	Sweet		
	Nutritional message	High energy density in food choice		
Knoeferle <i>et al.</i> (2015)	Low roughness	Sweet		
	Legato	Sweet		
	High pitch	Sweet		
	Low roughness	Sweet		
	Legato	Sweet		
	High pitch	Sweet		
	Low sharpness	Sweet		
	Consonant	Sweet		
	High roughness	Sour		
	High tempo	Sour		
	High sharpness	Sour		
	High pitch	Sour		
	Staccato	Sour		
	High sharpness	Sour		
	High pitch	Sour		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Kontukoski <i>et al.</i> (2015)	Staccato	Sour		
	High roughness	Sour		
	High roughness	Bitter		
	Staccato	Bitter		
	Low tempo	Bitter		
	Low pitch	Bitter		
	High roughness	Bitter		
	Legato	Bitter		
	Low sharpness	Bitter		
	Dissonant	Bitter		
	Low pitch	Salty		
	Staccato	Salty		
	High roughness	Salty		
	Low sharpness	Salty		
	Consonant	Salty		
	Long duration	Sweet		
	Consonant	Sweet		
	Legato	Sweet		
	Soft	Sweet		
	Low pitch	Sweet		
	Low brightness	Sweet		
	Low roughness	Sweet		
Wang and Spence (2015a)	High pitch	Sour		
	Dissonant	Sour		
	Long duration	Sour		
	High brightness	Sour		
	High roughness	Sour		
	High tempo	Acidity		
	Piano	Acidity		
	High pitch	Acidity		
	Low tempo	Low acidity		
	Piano	Low acidity		
	Cello	Low acidity		
	Legato	Low acidity		
	Consonant	Low acidity		
	Low tempo	Fruity		
	Piano	Fruity		
	Cello	Fruity		
	Legato	Fruity		
	Consonant	Fruity		
Wang and Spence (2015b)	High pitch		Vodka dislike	
	High roughness		Vodka dislike	
	Staccato		Vodka dislike	

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
	High tempo		Vodka dislike	
	High sharpness		Vodka dislike	
	High pitch		Citron vodka preference	
	High roughness		Citron vodka preference	
	Staccato		Citron vodka preference	
	High tempo		Citron vodka preference	
	High sharpness		Citron vodka preference	
	High pitch	Fruity		
	High roughness	Fruity		
	Staccato	Fruity		
	High tempo	Fruity		
	High sharpness	Fruity		
	Low roughness	Fruity		
	Legato	Fruity		
	Low sharpness	Fruity		
	High pitch		Raspberry vodka preference	
	Low roughness		Raspberry vodka preference	
	Legato		Raspberry vodka preference	
	Low sharpness		Raspberry vodka preference	
	High pitch		Vodka preference	
	Low roughness		Vodka preference	
	Legato		Vodka preference	
	Low sharpness		Vodka preference	
Wang <i>et al.</i> (2015)	Synthesizer	Sour		
	High pitch	Sour		
	Dissonant	Sour		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
	Legato	Sour		
	Piccolo	Sour		
	Clarinet	Sour		
	Staccato	Sour		
	Drums	Sour		
	Brass	Sour		
	Bass	Sour		
	Synthesized wave function	Bitter		
	Mid pitch	Bitter		
	Consonant	Bitter		
	Staccato	Bitter		
	Trombone	Bitter		
	Low pitch	Bitter		
	Legato	Bitter		
	Marimba percussion	Salty		
	Mid pitch	Salty		
	Consonant	Salty		
	Staccato	Salty		
	Synthesized bass	Salty		
	Pads	Salty		
	Piano	Salty		
	Low pitch	Salty		
	Saltshaker	Salty		
	Tearing paper	Salty		
	Bass	Salty		
	Brass	Salty		
	Dissonant	Salty		
	Synthesized bass	Sweet		
	Pads	Sweet		
	Piano	Sweet		
	High pitch	Sweet		
	Consonant	Sweet		
	Legato	Sweet		
	Strings	Sweet		
	Mid pitch	Sweet		
	Bells	Sweet		
	Synthesized piano reverb	Sweet		
	Wave functions	Sweet		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Yan and Dando (2015)	Airplane cabin noise Airplane cabin noise	Low sweet Umami		
Endo <i>et al.</i> (2016)	Crunchy chewing sound	Chewy		Crunchy chewing sound — interesting texture
Kantono <i>et al.</i> (2016)	Music preference Music dislike Music preference	Pleasant Bitter Sweet		
Reinoso-Carvalho <i>et al.</i> (2016)	High spectral balance Staccato Syncopated rhythm High pitch	Alcohol Alcohol Alcohol		
Wang and Spence (2016)	Dissonant Consonant Consonant	Sour Sweet Apple elderflower		
Endo <i>et al.</i> (2017)	Chewing sound Chewing sound Chewing sound Chewing sound Chewing sound Chewing sound Chewing sound Chewing sound Chewing sound Chewing sound	Expensive Fresh Palatable Hard Moist Rare Chewy Rough Pleasant Minced		Chewing sound — interesting / good texture / exciting / satisfying / appetite / comfortable of minced food / more ingredients perception of minced food / chewy of minced food
Guetta <i>et al.</i> (2017)	Consonant Low tempo Soft Violin	Sweet Sweet Sweet		
Reinoso-Carvalho <i>et al.</i> (2017)	Ascending scale Flute Legato Large hall reverberation Ascending scale Flute	Creamy Creamy Creamy Creamy	 Chocolate preference Chocolate preference	

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Spence and Wang (2017)	Legato		Chocolate preference	
	Large hall reverberation		Chocolate preference	
	Cork sound		Wine quality	
Wang and Spence (2017)	High pitch	Cold beverage		
	High tempo	Cold beverage		
Wang <i>et al.</i> (2017a)	High pitch	Spiciness of hot spicy food		
	High tempo	Spiciness of hot spicy food		
	Distorted timbre	Spiciness of hot spicy food		
	High pitch	Spiciness of medium spicy food		
	High tempo	Spiciness of medium spicy food		
	Distorted timbre	Spiciness of medium spicy food		
	High pitch	Spiciness of spicy–sweet food.		
	High tempo	Spiciness of spicy–sweet food.		
	Distorted timbre	Spiciness of spicy–sweet food.		
Wang <i>et al.</i> (2017b)				Sweet sour soundtrack — low sweetness / high sourness, Sour sweet soundtrack — high sweetness / low sourness

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Kantono <i>et al.</i> (2018)	Music preference	Sweet		Neutral (in preference) music — milkiness / cocoaness
Rahne <i>et al.</i> (2018)	White noise	Low sour		
	White noise	Sweet		
Roque <i>et al.</i> (2018)	Ice cube sound	Fresh		
	Carbonation sound	Fresh		
Sinesio <i>et al.</i> (2018)	Nature		Tomato preference	
	Nature		Wild rocket preference	
	Nature	Freshness of vegetable		
Wang and Spence (2018b)	Bells		Wine preference	
	Piano		Wine preference	
	Synthesizer		Wine preference	
	Consonant		Wine preference	
	Legato		Wine preference	
	Piccolo		Wine dislike	
	Clarinet		Wine dislike	
	Dissonant		Wine dislike	
	Staccato		Wine dislike	
	Bells	Sweetness of a sweet wine		
	Piano	Sweetness of a sweet wine		
	Synthesizer	Sweetness of a sweet wine		
	Consonant	Sweetness of a sweet wine		
	Legato	Sweetness of a sweet wine		
	Piccolo	Acidity		
	Clarinet	Acidity		
	Dissonant	Acidity		
	Staccato	Acidity		
	Sparse texture	Body		
	Staccato	Body		
	Sparse texture	Balance		
	Staccato	Balance		
	Sparse texture	Length		
	Staccato	Length		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Wang and Spence (2018c)	Sparse texture	Acidity	Wine preference	
	Staccato		Wine preference	
	Consonant			
	Consonant	Sweet	Juice preference	
Ziv (2018)	Pleasant	Acidity		Pleasant music — prestigious
	Pleasant	Acidity		
	Pleasant	Acidity		
	Pleasant	Body		
	Pleasant	Soft		
Ba and Kang (2019)	Conversation	Olfactory intensity	Olfactory discomfort	Low volume — music–aroma matching
	Traffic		Olfactory discomfort	
	Urban		Olfactory discomfort	
	Urban		Olfactory discomfort	
	Birdsong		Lilac matching	
	Conversation		Lilac matching	
	Birdsong		Coffee matching	
	Conversation		Coffee matching	
	Birdsong		Bread matching	
	Conversation		Bread matching	
	Birdsong		Osmanthus matching	
	Birdsong		Osmanthus matching	
Burzynska <i>et al.</i> (2019)	Low pitch	Wine aroma intensity		
	Low pitch	Wine full body		
Cho <i>et al.</i> (2019)				Downtempo music — preference of food in blue plate, Uptempo music — preference of food in red plate
Fiegel <i>et al.</i> (2019)	Low pitch	Chocolate texture preference		Music — bell pepper texture preference, Music — texture preference/ flavor intensity
	Low pitch	Bell pepper preference		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Hauck and Hecht (2019)	High tempo	Bell pepper flavor		
	High volume	pleasant		
	High volume	Chocolate texture preference		
	Low volume	Chocolate flavor pleasant		
	Pleasant Music preference	Bell pepper intensity Pleasant Beverage like		Harmonic/mellow/light/ fruity/salty/foul/powerful/ sour/gloomy music – beverage harmonicity/ mellowness/lightness/ fruitiness/saltiness/foulness/ power/sourness/gloominess
Höchen-berger and Ohla (2019)	Sound	Toffee intensity		
	Low pitch	Bitter		
	Trombone	Bitter		
	High pitch	Sweet		
Hsu and Chen (2019)	Piano	Sweet		
				French music — French wine selection,
				Spanish music — Spanish wine selection.
	Music preference	Sweet		Neutral (in preference)
	Music preference	Milky		music — cocoaness at the beginning,
Kantono <i>et al.</i> (2019)	Music dislike	Bitter		Neutral (in preference)
	Music dislike	Creamy		music — bitterness,
	Music dislike	Vanilla		Neutral (in preference)
				music — milkiness/roasted/ cocoaness.
Lin <i>et al.</i> (2019)	Café environment	Sweet		
	Park environment	Sweet		
	Bar environment	Bitter		
	Food court environment	Bitter		
	Fast food environment	Bitter		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Luca <i>et al.</i> (2019)	Bar environment	Milky		
	Fast food environment	Milky		
	Café environment	Creamy		
	Park environment	Creamy		
	Food court environment	Cocoa		
	Café environment	Roasted		
	Bar environment	Roasted		
	Food court environment	Roasted		
	Fast food environment	Roasted		
	Bar environment	Cocoa		
	Fast food environment	Cocoa		
	Classical	Delicateness of white wine		
	Classical	Sweetness of white wine		
	Popular	Alcohol of red wine		
Mitchell <i>et al.</i> (2019)	Effervescence sound	Low astringency		
	Effervescence sound	Low bitter		
	Effervescence sound	Carbonation		
Motoki <i>et al.</i> (2019)	Low pitch voice	Sour		
Reinoso-Carvalho <i>et al.</i> (2019)	Negative music	Alcohol		
	Positive music	Beer		
	Positive music	Sweet		
	Negative music	Bitter		
	Negative music	Alcohol		
	Positive music	Body		
	Negative music	Body		
	Negative music	Body		
	Positive music	Beer		
	Positive music	Sweet		
	Negative music	Bitter		
Wang <i>et al.</i> (2019a)	Cello	Fruity		
	Woodblock	Fruity		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Wang <i>et al.</i> (2019b)	Drone	Fruity		
	Low pitch	Fruity		
	Cello	Smooth		
	Woodblock	Smooth		
	Drone	Smooth		
	Low pitch	Smooth		
	Low tempo	Bitter		
	Sound intensity	Low astringency		
	Low spectral centroid	Low astringency		
	High tempo	Low astringency		
	Consonant	Low astringency		
	Dissonant	Low astringency		
	Sound intensity	Bitter		
	Low spectral centroid	Bitter		
	High tempo	Bitter		
	Consonant	Bitter		
Wang <i>et al.</i> (2019c)	Dissonant	Bitter		
	High spectral centroid	Acidity		
	Bells	Sweet		
	Piano	Sweet		
	Synthesizer	Sweet		
	Consonant	Sweet		
	Legato	Sweet		
	Low pitch	Low sweet		
	Synthesized bass	Low sweet		
	Consonant	Low sweet		
Xu <i>et al.</i> (2019)	Legato	Low sweet		
	Café environment	Bitter		
	High sharpness (birds)	Creamy		
	High sharpness (forest)	Sweet		
	Machine sound	Bitter		
	Birdsong	Creamy		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Bravo-Moncayo <i>et al.</i> (2020)	Filtered noise Filtered noise Filtered noise	Bitter Olfactory intensity High temperature		
Covaci <i>et al.</i> (2020)				Atmospheric audio — olfactory annoyance, Visual–congruent audio — olfactory distracting perception, Visual–congruent audio — olfactory annoyance
Fibrianto <i>et al.</i> (2020)	Low tempo High tempo Low tempo High tempo Low tempo High tempo Low tempo High tempo Long duration Long duration	Low sour Low sour Bitter Bitter Astringency Astringency Tooth etch Tooth etch Low bitter Sweet		
Nivedhan <i>et al.</i> (2020)	Positive music Positive music	Sweet Coffee preference		
Peng-Li <i>et al.</i> (2020)	High pitch Legato Consonant Low tempo Warm timbre Clarinet Minor mode Staccato High tempo	Sweet Sweet Sweet Sweet Sweet Sweet Salty Salty Salty		
Qi <i>et al.</i> (2020)	Low pitch Mid pitch Yunluo Guzheng Dizi Erhu Yunluo	Bitter Salty Sweet Sweet Bitter Bitter Umami		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Reinoso-Carvalho <i>et al.</i> (2020a)	Positive music		Chocolate preference	
	Soft		Chocolate preference	
	Positive music	Sweet		
	Positive music	Soft		
	Negative music	Bitter		
Reinoso-Carvalho <i>et al.</i> (2020b)	Soft		Chocolate preference	
	Soft	Soft		
	Soft	Chocolate intensity		
	Positive music		Chocolate preference	
	Positive music	Sweet		
Wang <i>et al.</i> (2020)	Positive music	Soft		
	High pitch	Sweet		
	Bells	Sweet		
	Piano	Sweet		
	Synthesizer	Sweet		
	Consonant	Sweet		
	Legato	Sweet		
	Low pitch	Bitter		
	Synthesized wave functions	Bitter		
	Consonant	Bitter		
	Legato	Bitter		
	Trombone	Bitter		
	High pitch	Sweet		
	Bells	Sweet		
	Piano	Sweet		
	Synthesizer	Sweet		
	Consonant	Sweet		
	Legato	Sweet		
	Low pitch	Bitter		
	Synthesized wave functions	Bitter		
	Consonant	Bitter		
	Legato	Bitter		
	Trombone	Bitter		
Padulo <i>et al.</i> (2021)	Mid pitch	Salty		
	Staccato	Salty		
	Dissonant	Salty		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Pantoja and Borges (2021)	High pitch	Sweet	Burger preference Pizza preference	
	Legato	Sweet		
	Consonant	Sweet		
	High tempo			
	High tempo			
Rodríguez <i>et al.</i> (2021)	Bottle pouring sound	Fresh		
	Bottle pouring sound	Fresh		
	High volume	Fresh		
	Low volume	Sweet		
	Low pitch	Bitter		
Speed <i>et al.</i> (2021)	High volume	Alcohol	Raspberry matching Menthol matching Lemon matching Onion matching	
	High pitch			
	High pitch			
	High pitch			
	Low pitch			
Wang <i>et al.</i> (2021)	Staccato	Salty		
	Harmonic complexity	Salty		
	Low randomness	Salty		
	High roughness	Salty		
	High tempo	Salty		
	Long attack	Salty		
	Long decay	Salty		
	Reverberation	Salty		
	Minor mode	Salty		
	Minor mode	Salty		
Lin <i>et al.</i> (2022)	Music preference	Sweet		
	Pleasant	Sweet		
	Music dislike	Bitter		
	Unpleasant	Bitter		
	Music preference	Cocoa		
	Pleasant	Cocoa		
	Music dislike	Roasted		
	Unpleasant	Roasted		
	Music dislike	Low sweet		
	Unpleasant	Low sweet		
Motoki <i>et al.</i> (2022)	Jazz	Healthy savory		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
Ramos-de-la-Peña <i>et al.</i> (2022)	Classical	Healthy savory		
	Classical	Low indulgent savory		
	Classical	Sweet		
				Heat evoking video sound — hotness / moisture / saltiness / low bitterness, Cold evoking video sound — sweetness, Sad evoking video sound — ketchup moisture / saltiness, Height / fear evoking video sound — ketchup oiliness, Pleasant / speed / fear / relaxation evoking video sound — ketchup creaminess / oiliness, Hot evoking video sound — salad dressing creaminess, Sad evoking video sound — salad dressing hotness / low creaminess, Cold evoking video sound — salad dressing saltiness / oiliness, Height / fear evoking video sound — salad dressing oiliness, Relaxation evoking video sound — oiliness / creaminess.
Vanden-berghe-Descamps <i>et al.</i> (2022)				Irish / USA soundtracks — beer harmony / familiarity / pleasantness Indian soundtrack — Irish beer USA soundtrack — low Irish beer
Guedes <i>et al.</i> (2023a)	Mid–high pitch Guitar Woodwind	Sweet Sweet Sweet		

Table A5.
(Continued)

Reference	Auditory	Perception	Personal preference	Other auditory–chemosensory associations
	Legato	Sweet		
	Low tempo	Sweet		
	Mid–high pitch		Sweet vegetable like	
	Guitar		Sweet vegetable like	
	Woodwind		Sweet vegetable like	
	Legato		Sweet vegetable like	
	Low tempo		Sweet vegetable like	
	Mid–high pitch		Nonsweet vegetable like	
	Guitar		Nonsweet vegetable like	
	Woodwind		Nonsweet vegetable like	
	Legato		Nonsweet vegetable like	
	Low tempo		Nonsweet vegetable like	
	Mid–high pitch		Sweet cookie like	
	Guitar		Sweet cookie like	
	Woodwind		Sweet cookie like	
	Legato		Sweet cookie like	
	Low tempo		Sweet cookie like	
	Mid–high pitch		Nonsweet cookie like	
	Guitar		Nonsweet cookie like	
	Woodwind		Nonsweet cookie like	
	Legato		Nonsweet cookie like	
	Low tempo		Nonsweet cookie like	